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Anti-sloshing Devices in Liquid Propellant Tanks for Space Vehicles

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Summary

As far as space vehicles are concerned, they need to follow a very accurate trajectory in order to enter a certain orbit and perform a specific mission. To this extent, the undesired oscillation of the liquid mass inside propellant tanks, commonly referred to as *sloshing*, can lead to stability and flight control issues. This could be avoided with the use of several baffles inside the tank and a simple mechanical model can be addressed to evaluate the damping provided by said devices.

The overall problem can be analyzed through a mass-pendulum-dashpot model which allows to predict the propellant sloshing through an equivalent moving mass attached to a damped pendulum. The empirical expressions used to evaluate the mechanical model parameters, i.e. slosh mass, slosh frequency, pendulum hinge point location and damping ratio, are validated through the use of test data acquired on a sub-scale tank. In particular, the tank used to collect data is a sub-scale version of the Ares I US LH2 tank. Eventually, a comparison between analysis and experimental results will be presented.

The aforementioned mechanical model will be implemented to evaluate the damping offered by four ring baffles in the Efesto liquid oxygen tank. Efesto is a bi-liquid propelled rocket designed by the PoliTo Rocket Team, a stutents team within the Politecnico di Torino. The purpose of this investigation is to validate the effectiveness and usability of such formulations, with an attempt to estimate the real damping the tank would experience with a four-baffle configuration.

Chapter 1

Anti-sloshing devices

1.1 Introduction

Among the countless challenges that engineers have to address when designing a space vehicle, the slosh phenomenon in the propellant tanks stands out as a critical issue to be aware of. Indeed, propellant motion is of great concern during the process of outlining flight control systems for liquid-propelled rockets as it largely affects the attitude stability and the dynamic control of the vehicle. This is even more noticeable in modern space launch vehicles as they have grown to massive scales and nearly 90% of their liftoff mass is purely liquid propellant. In particular, the launch vehicle suffers the longitudinal load produced by both the engine thrust and the aerodynamic drag. To make things worse, the flight attitude of the vehicle is always changing, producing vibration loads which are transmitted from the structure to the liquid in the tanks, leading to the sloshing effect.

Another field of interest is the detumbling process performed by a service satellite to stop the rotation of space debris, as the majority of them are tumbling due to their residual angular momentum. During this procedure a jet detumbling payload is rigidly attached to the target, resulting in a forceful impact. It is clear that the accelerations involved are not negligible and are worth considering for the propellant sloshing.[1]

Other than stability and controllability issues, the propellant motion can eventually lead to unwanted aerolastic effects. Launch vehicles are long and slender bodies and therefore resemble a flexible beam, with the consequence of plausible coupling between vibration characteristics of liquid and container. Indeed, results demonstrate that the elastic deformation and liquid sloshing change the motion characteristics of the rocket and is something to be aware in the design phase.[2]

The sloshing phenomenon can be defined as the oscillation of the free surface of a liquid inside a partially filled container. The behaviour of the oscillations depends on many factors, such as the vehicle dynamics, container geometry, fluid properties, fluid fill level and the damping of the system. The slosh dynamics requires, then, a thorough analysis

to understand how the fluid motion affects the vehicle stability along its path. A detailed evaluation of this phenomenon can be performed through experimental method, analytical theoretical method and numerical simulation. The latter would require extensive CFD calculations, as it turns out to be an extremely powerful tool to predict real motion of the fluid. The drawback of such approach is a too heavy workload required to solve ordinary design analyses, especially in the early-stage. Thereby a lighter and simpler approach is often preferred and it features the replacement of the moving liquid in the container with an equivalent mechanical model.

The most widely adopted models consist of either a mass-pendulum-dashpot system or a damped mass-spring system which are designed in such a way that they have the same resultant moving mass, damping and frequency as the actual system. The parameters required by these models include natural frequency of the sloshing wave, slosh mass, slosh mass center location (i.e. pendulum hinge location) and damping ratio. All of these are required to represent *linear regime slosh*, where the wave amplitude is not very high and the liquid dynamic characteristics do not alter with wave amplitude. If the wave amplitude were to be too high, then liquid surface breakup and other complex fluid motion would develop and nonlinear damping models should be accounted for [3].

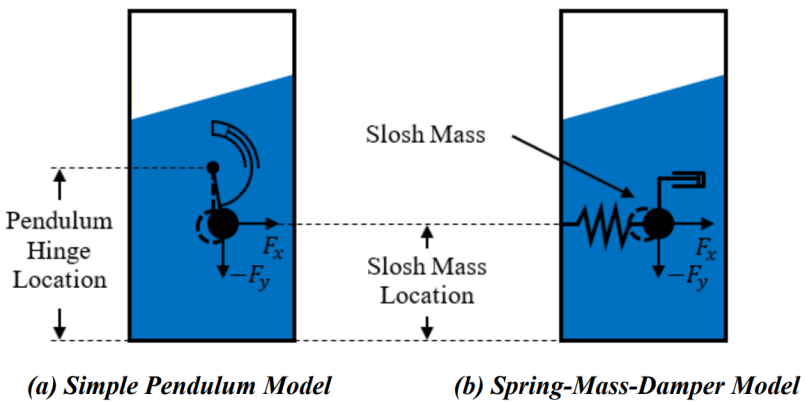


Figure 1.1: Simple mechanical models representing liquid slosh

Both of these models split the liquid mass into two or more parts in order to preserve the bulk force and moments on the tank. The portion of the equivalent liquid mass that is not moving is represented as a rigid-body mass inside the tank (not shown in the above pictures). All the other portions, instead, are considered as sloshing masses connected to a damped pendulum system or a damped spring system. The number of slosh masses is given by the total modes of the linear system, although for this report only the first mode will be evaluated since higher order modes have a negligible effect. Once the system parameters are calculated and the analysis is performed, minimum damping requirements for stable flight control are issued [4].

It is worth stating that such linear representation of a complex three-dimensional problem

carries along some limitations to the model. These are primarily the unsuitability of the model near the natural frequencies of the propellant modes, i.e. at resonance, and the impossibility to predict the so-called *rotary slosh* [5]. The latter comes as a result of the coupling between the degrees of freedom of the lateral modes, which cause the fluid to start rotating while oscillating. It presents a risk to flight control as it involves the formation of a stable limit cycle which can lead to undesirable roll coupling.
Indeed, experimental data recorded by Hutton [6] showed that the fluid motion inside a cylindrical container, under constant sinusoidal excitation, can be split up in four response regions according to the input frequency.
Starting off, liquid experiences stable and in-plane motion for below-resonance frequencies, then undergoing chaotic motion near resonance. Just above resonance rotary motion finds his way in a narrow band and it can either become stable rotary or again planar motion with an additional increase in frequency. However, the mechanical model used to predict rotary sloshing further complicates the analysis and therefore will not be discussed in the paper.

To reduce the effects of liquid sloshing, some devices can be installed inside the container with the main purpose of suppressing the oscillations. The aforementioned devices are typically anti-sloshing baffles with various geometries which help dissipate the energy of the fluid. The first slosh damping mechanism is actually found in nature and does not need any implementation: it is the viscosity of the liquid itself which smooths out velocity gradients by definition. This effect is more evident within a thin layer on the inner wall of the tank, where the so-called Stokes boundary layer is found. If the tank is bare-walled, meaning it does not have any internal hardware, then the *smooth-wall damping* should be considered. With the addition of anti-sloshing baffles the boundary of the fluid is changed and a stronger damping takes place, as the baffle hinders the flow of the liquid. This creates a pressure differential across the baffle, eventually resulting in passing flow resistance and subsequent reduction of the liquid sloshing [1].

1.2 Anti-sloshing baffle design

The damping requirement can be met through many different configurations, which have evolved through the years, allowing for a wider range of options to pick from in the design phase. All of them share the same need to prevent instabilities in the control system and keep the vehicle at the desired target attitude.

As a matter of fact, since the early NASA programs, as the Apollo, through the Space Shuttle and up to the younger Ares or Space Launch System (SLS) it has been necessary to verify the stable flight control damping requirements. They involve the validation of the slosh mechanical model as well as investigate the influence of the baffles on propellant motion. Several experiments on sub-scale cylindrical tanks, involving lateral slosh excitation, have been extensively carried out in the past and are collected in a milestone book written by Dr. Norm Abramson in 1966 [7]. It contains a very comprehensive knowledge on both experimental and analytical results of sloshing, with the latter being derived for

simple geometries such as rectangular boxes or the more common flat-bottomed cylindrical tank. It is worth pointing out that for this geometry Miles, in 1958, managed to obtain an analytical expression for the free surface damping produced by a flat plate, using data from experiments of Keulegan and Carpenter [8]. This formulation is more widely known as the *Miles equation* and will be discussed in the next paragraphs.
These studies proved crucial for the development of the analysis tools used in the slosh dynamics, which consist of the slosh mechanical model and slosh damping expressions we still use today for an early stage design phase.

When baffles are installed in the inner wall of the tank, the dissipation of the Stokes boundary layer and the dissipation due to viscosity become negligible in the proximity of the device itself.
A general equation allows to calculate the energy dispersed due to the pressure difference resistance offered by the baffle:

$$\begin{cases} D_d = \rho \int_{S_w} C_d |\nabla\varphi \cdot \vec{n}|^3 dS \\ C_d = \frac{K}{\sqrt{\eta_{\max}}} \end{cases} \tag{1.1}$$

Where $\vec{n}$ is the unit normal vector of the designed anti-sloshing baffle, $\eta_{\max}$ is the maximum wave amplitude, K is related to the curvature of the baffle and varies upon the configuration. Finally, the C_d value is limited by the Reynolds number.
The value of the drag coefficient for free flat plates in oscillating liquids can assume a more precise formulation [8], as follows:

$$C_d = 15 \left(\frac{U_m T}{D} \right)^{-1/2} \tag{1.2}$$

Where U_m is the time-wise maximum velocity, T the period and D the plate width; it is evident that the drag coefficient is here dependent on the period parameter.

The total mechanical energy of the liquid sloshing is:

$$E = \frac{\rho}{2} \int_{\Omega} |\nabla\varphi|^2 d\Omega \tag{1.3}$$

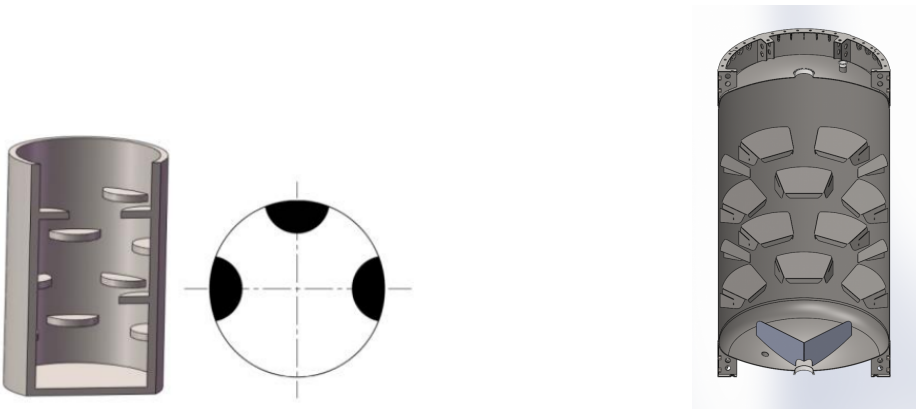
And it allows to have a general expression for the damping ratio:

$$\zeta = \frac{D_d}{2\omega E} \tag{1.4}$$

These formulations will be further developed according to the baffle design and the operating conditions. Two main distinctions of all possible configurations are to be made according to the working principle behind the damping mechanism. There are passive devices, such as the the well-known baffles which are mounted in the tank and help dissipate as much energy as possible, or active devices, such as thruster firings which counteract the fluid forces by applying an opposite and equal force or momentum.
The passive damping devices group several structures, including semicircular baffles, radial baffles, annular ring baffles, perforated structures and floating baffles. A brief description of these setups is proposed in the next paragraphs [1].

1.2.1 Semicircular baffles

This is a type of asymmetric baffle design, as shown in figure 1.2a. It is made up of several layers of semi-circular shaped baffles at different heights inside the tank. It has one main constraint, which is determined by the angle between the central axis and the liquid sloshing direction; if it holds a value greater than 60° then the baffle becomes ineffective.



(a) General design (b) Copenhagen Suborbitals tank section view

Figure 1.2: Semicircular baffle

As the image 1.2b shows this is the layout adopted for the liquid oxygen tank onboard the Spica I rocket by Copenhagen Suborbitals, a Danish rocketry company. Such plates are installed with a very clever expedient, that is a tiny slot along the side of the baffle support, which allows the tank to deform and expand.

In theory, the bigger the radius of the semicircular baffle r, the higher the damping efficiency, even though efficiency itself is not completely correlated with the radius alone. The bearing capacity is not excellent and the stiffness of the baffle is weaker than other configurations, thus resulting in a relatively short plate (typically around $0.2R$ or $0.3R$). The dimension of the baffle comes as a result of the balanced compromise between the above mentioned parameters and hence is not unique.

There is an empirical formula which estimates the damping of this type of baffle, which is:

$$\zeta = KC\left(\frac{d}{R,\theta}\right)\alpha^k\left(\frac{\eta}{R}\right)^{0.5} \tag{1.5}$$

In this formula, k and K are both coefficients, whose values are $2.83 \sim 3$ and $1.5 \sim 1.8$. The term α refers to the ratio of the sum of the area of all semicircular baffles to the tank cross-sectional area: $\alpha = \sum_i A_i/\pi R^2$, whereas $C(\frac{d}{R,\theta})$ does not have a fixed expression as is obtained by fitting the data.

1.2.2 Radial baffles

Radial baffles are also known as vertical strip baffles and are evenly separated along the circumference of the tank and they develop in a bottom-up way, up to a fixed height h. What makes this type of baffle stand out is the efficacy of the damping regardless of the liquid fill level. Indeed, since they run along the wall of the tank starting from the bottom, the attenuation effect remains unchanged as the propellant is being consumed with the liquid level decreasing accordingly. However, at the beginning of the mission when the tanks are filled up, the liquid free surface may outstrip the end of the baffle, in case it does not fully extend to the top, leading to insignificant damping. Furthermore, they have a better damping efficiency when the oscillation is non-planar, meaning some rotary sloshing is occurring.

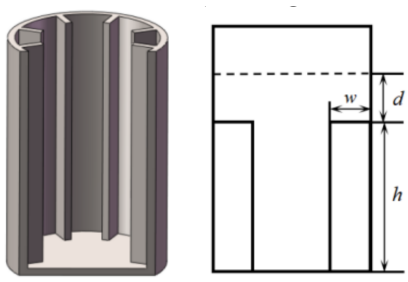


Figure 1.3: Radial baffle

The design parameters include the baffle width w, the height h and the number of plates installed n (i.e. the angle distance between two adjacent baffles).
If the number of baffles increases, the anti-sloshing effect will also increase; the drawback will be a remarkable mass increase. Also, the directionality of the damping will be affected and should be accounted for when designing the tank.
Again, the damping ratio can be calculated through an empirical formula, which is specific for this design:

$$\begin{cases} \zeta = \frac{0.13}{1+\kappa}\left(\frac{w}{R}\right)^{1.5}\left(1-\frac{w}{2R}\right)^{-2.5}\left(\sum_{i=1}^{n}|\cos\theta_i|^{2.5}\right)\left(\frac{\eta}{R}\right)^{0.5} \\ \kappa = 0.209\left(\frac{w}{R}\right)^{2}\left(1-\frac{w}{2R}\right)^{-2}\left(\sum_{i=1}^{n}\cos^2\theta_i\right) \end{cases} \tag{1.6}$$

Where η is the shaking width of the storage tank, θ_i is the angle between the i-th radial baffle and the direction of liquid sloshing. This is, as anticipated, a function of the number of baffles.

1.2.3 Annular ring baffles

The flat annular ring baffle is essentially a circular ring parallel to the tank cross-sectional area and is installed at a specific height H from the bottom of the tank. The tank now becomes axisymmetric, resulting in the absence of a specific direction along which damping occurs, since it now is independent of the slosh direction.

The radial plates are installed on the bare-walled tanks and have a thickness such that they can be modeled as rigid bodies. More specifically, the flexibility parameter F is what determines the behavior of the baffle. When this parameter is equal to zero the baffle can be considered as rigid, thus enhancing the damping and anti-shake characteristics of the device.
The specific calculation formula for the baffle flexibility is as follows:

$$F = \frac{\rho g R(1-\nu^2)}{E}\left(\frac{w}{R}\right)^5\left(\frac{R}{\tau}\right)^3 f\left(\frac{w}{R}\right) \tag{1.7}$$

where ρ is the liquid density, R the radius of the tank, ν the Poisson's coefficient of the baffle material and E the Young's modulus of said material. Geometry-wise w stands for the width of the baffle, whereas τ its thickness. Finally, $f(w/R)$ is a function which describes the effect of boundary conditions on the baffle.

When it comes to the design parameters, they are the width of the baffle w, the layer spacing s (i.e. the spacing between adjacent baffles) and the number of baffles. In fact, it may be convenient to not have a single-layer baffle but to opt for multiple layers to benefit an increased anti-sloshing effect but mostly to account for the decrease of liquid level as it gets burnt out from the engine.

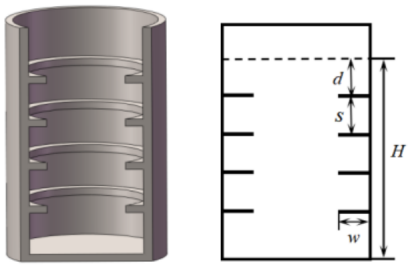


Figure 1.4: Annular baffle

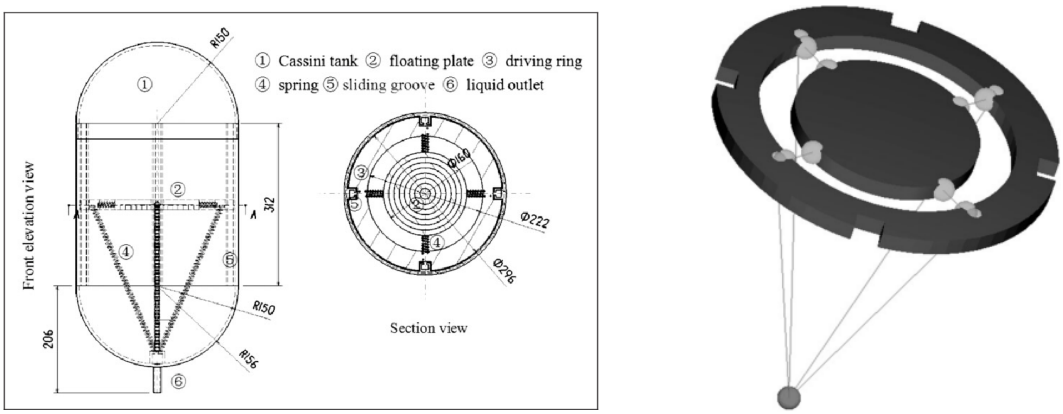
As figure 1.4 shows there is another parameter, d, which gives information about the distance between the liquid level and the closest annular baffle. If this distance happens to be negative, it means that the baffle exceeds the liquid level and the beneath-laying baffle needs to be considered for the damping mechanism.
The range of d which makes the baffle effective is $-0.1R \leq d \leq 0.45R$. However, when estimating the damping of the tank, the bottom baffle plays almost no role unless there is very little liquid left in the reservoir.
When it comes to spacing, the optimal distancing between the baffles is designed as $0.2R \leq s \leq 0.4R$ to make the overall damping relatively uniform.

The damping offered by such device is thoroughly analyzed in the dedicated Anti-slosh flat ring baffle section of the paper.

1.2.4 Floating baffle

Another type of slosh-suppression device is a so-called *drive ring*, which breaks the traditional idea of installing a fixed baffle in the tank. It works by drastically reducing both the sloshing force and the liquid sloshing kinetic energy.
This structure consists of a floating plate, a transmission ring and a buffer spring. The floating baffle is the core element which allows the damping to occur and is connected through four pre-stressed springs to the transmission ring, as figure 1.5b illustrates.
It has also four vertical chutes on the tank wall which limit the movement of the transmission ring along the longitudinal axis of the tank only, accounting for the propellant leaving the tank as it is being used up. This carries along the drawback of having moving parts inside the tank with a considerable increase in the risk of breakup of some elements when subjected to heavy sloshing loads.

This design has been taken into account for the Cassini tank [9]. Cassini is a NASA's spacecraft targeting the exploration of Saturn and its family of icy moons, which has been operative for twenty years before plunging and collapsing into the Saturn's atmosphere.



(a) Baffle setup

(b) Baffle and transmission ring

Figure 1.5: Floating baffle in the Cassini tank

This is a new approach to slosh damping in space vehicle's tanks as it has been found to be a much lighter option compared to fixed baffles, with a further increase in damping characteristics. What stands out is the baffle capacity to move with six degrees of freedom under the impact of the liquid, therefore softening the slosh wave independently of its direction.
However, due to its working principle, no analytical expression has been developed yet and the dynamic data generated by the liquid sloshing and relative damping can only be obtained by numerical simulation.

1.2.5 Perforated damping structure

The perforated damping structure is another viable option for the slosh suppression in a large-capacity propellant tank. It consists of a conical perforated type PMD, Propellant Management Device [10]. This design has been tested to simulate sloshing characteristics in a satellite propellant tank and how it affects the attitude control of the satellite in orbit.

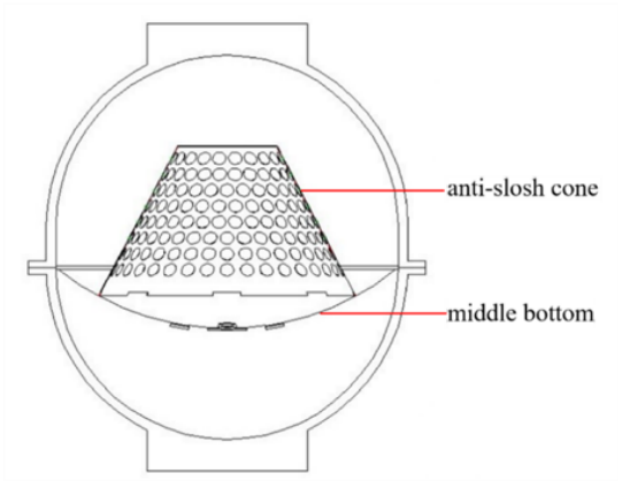


Figure 1.6: Perforated damping structure

The device consists of a middle bottom and an anti-slosh cone with mesh holes (fig. 1.6). The middle bottom splits the tank into two regions, the upper tank and the lower tank, with test liquid only circulating across the mesh screen when moving from one part to the other. This is how the damping occurs, namely it relies on the energy dissipation yielded by the anti-slosh cone structure.

1.2.6 Active slosh control

The anti-sloshing effect is achieved by applying a transverse force or a pitching moment on the vehicle through control effectors, e.g. gas jet thrusters. They would be activated in the case of propellant moving inside the tanks in order to always ensure a relative equilibrium between the vehicle and the fuel slosh dynamics [11]. In particular, a similar slosh suppression method is installed onboard the Ariane 6 upper liquid propulsion module (ULPM), where the guidance, navigation and control (GNC) algorithm commands thruster firings. This approach is required to keep the upper stage at the target attitude when performing settling maneuvers where thruster fire and sloshing occur at a relatively low frequency. [12].

1.3 Anti-slosh flat ring baffle

Among all the above analyzed designs, the most widely used one corresponds to the flat annular baffle as it shows very good accordance with analytical damping predictions. Furthermore, meticulous and in-depth experiments concerning this baffle geometry are found on many papers, which represent a plentiful source to draw data from when validating the semi-empirical expressions.
To this extent, a deeper investigation on this design is presented in the current section and a comparison between experimental and analytical results is proposed in chapter 2.

In the design of anti-slosh ring baffles, the most widely used damping equation is the one obtained empirically by Miles, which is based on the experiments of Keulegan and Carpenter on fluid drag in flat plates [8]. This equation calculates the damping of a single-layer baffle and presents itself in several forms. The most generic one for cylindrical tanks is [4]:

$$\zeta_{Miles} = \frac{15(4/3\pi)^2 \alpha A e^{-4.601 d/R} \sqrt{\eta w}}{2\sqrt{\pi}\,(m_s/\rho)\,\Gamma^2} \tag{1.8}$$

where ζ_{Miles} is the baffle damping, d is the depth of the baffle below the undisturbed fluid free surface, A is the tank cross-sectional area, η is the slosh wave amplitude at the tank wall (measured with respect to the undisturbed free surface), w is the baffle width, ρ the propellant density, Γ the ratio of the lateral displacement of slosh mass to slosh wave amplitude and α is the fraction of tank cross-sectional area occupied by the baffle. With a little geometry calculation the value of α can be obtained from the following:

$$\alpha = \frac{2w}{R} - \frac{w^2}{R^2} \tag{1.9}$$

The reader should refer to figure 1.4 for a visual understanding of the design.

Equation 1.8 can be further simplified for a circular cylindrical tank with flat annular horizontal ring baffles. Under some hypothesis as a relatively small radial length of the baffle and a relative liquid height of $h/R > 2$, the parameters of the above equation become [13]:

$$m_s = 0.456(\rho A R) \qquad f_b = e^{-1.84(h_s/R)} \qquad \Gamma = 0.649 \tag{1.10}$$

By substituting the parameters 1.10 into the general form of Miles equation 1.8 it is possible to obtain a much more straightforward and easy-to-use formulation for the damping ratio [8]:

$$\zeta_{Miles} = 2.83 e^{-4.601 d/R} \alpha^{1.5} \sqrt{\frac{\eta}{R}} \tag{1.11}$$

This analytical expression is much simpler than the previous one, making it preferable especially in the design of more sophisticated anti-sloshing baffle systems.

The very powerful tool that accompanies this formulation and the mechanical model is the linear superposition principle. It allows to approximate the damping of multi-layer baffles as the superposition of damping of multiple single-layer baffles. This is valid as long as the spacing between baffles does not drop below one baffle width.

Extensive experiments have been carried out on sub-scale model tanks to extract valuable data regarding the mechanical model parameters and to validate the accuracy of all semi-empirical expressions used in the model. Therefore in this paper some test data recorded at the NASA Marshall Space Flight Center and collected in a paper by Perez et al. [4] will be used as reference to draw a comparison between analytical and experimental results in section 2.2. In the above mentioned paper all the parameters are given in a dimensionless form, since dimensional analysis shows that slosh masses are scaled in proportion to the total liquid mass and all length dimensions are scaled geometrically.
For a geometrically scaled tank, one of the most important scaling parameters is the Froude number, which shows that all frequencies are scaled in proportion to $g^{0.5}/R^{0.5}$. Similarly, it can be observed that smooth-wall damping, i.e. the damping of bare-walled tanks, is proportional to the factor $\nu^{0.5}/(R^{0.75}g^{0.25})$, similar to Galilei number or Reynolds number. When it comes to baffle damping, it generally scales one-to-one in terms of logarithmic decrement between sub-scale and full-scale models, meaning no scaling factor is necessary. In the latter case geometrically scaled baffles and slosh wave amplitudes are implied.

Slosh experiments have been performed on 1/5 scale Ares I US LOX and LH2 tank models with the purpose of validating analysis tools used to predict mechanical model parameters as well as estimate the damping provided by annular baffles under several configurations. Tests have been performed with multiple baffles with varying fluid fill levels to determine the damping effect. A picture of the liquid hydrogen tank is shown in figure 1.7.
Two different test methods were used, in particular Random Excitation (RE) testing was used in cases in which slosh mass was of interest, whereas Free Decay (FD) testing was adopted to determine nonlinear behaviour and damping dependency on slosh wave amplitude.

Annular baffles are excellent at damping along the radial velocity direction, but struggle to dissipate energy for motion tangent to the wall. This tells us that the damping mechanisms here discussed do not provide a general damping, but rather are strictly dependent on the in-plane motion of the slosh wave. As previously announced there is the possibility of the development of rotary slosh inside the container with a constant wave amplitude and in such a case the baffles efficiency would drop (figure 1.8).

This phenomenon could be caused by two main factors, both sharing the same principle, that is the preferred orientation of the slosh, which do not always correspond to the targeted direction. The first element causing the "swirl" motion could be a slightly out-of-roundness of the tank. The other potential cause of swirling could be the filling process. Indeed, in this phase a circular flow is very likely to develop in the tank and if this flow is not allowed to completely die out before testing, then it will negatively affect test results. More specifically, the damping values estimated with swirl will appear higher than if damping values were estimated in a tank without rotary motion.

Figure 1.7: Ares I US LH2 tank model with flat ring baffles

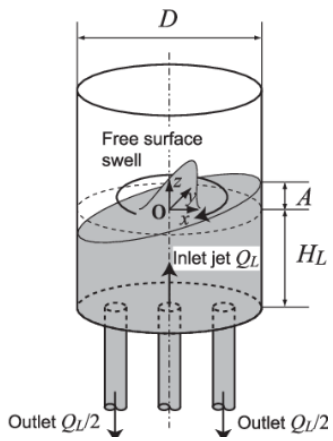


Figure 1.8: Rotary sloshing example

In this event, the baffle will not allow to dissipate energy at the same rate as a purely-radial slosh wave would. The slosh wave would start rotating off of the excitation axis, bringing to a noticeable decrease in liquid kinetic energy. To account for this, the mechanical model can be enriched by incorporating the *split damping* [5], where a smooth-wall viscous damping value is applied to the tangential component of the sloshing velocity but a baffle damping value is applied to the radial component.

Chapter 2

Validation of Miles equation

2.1 Mechanical model

As above stated, the sloshing phenomenon can be represented by a mechanical model quite accurately, with the advantages of simplifying the mathematical treatment and quantifying the damping provided by anti-sloshing devices. Indeed, in this paper the equivalent mass-pendulum-dashpot mechanical model will be discussed, where the model parameters are obtained through empirical formulas. Most of these have been obtained by Abramson in 1966 [7] but are still widely used in more recent early-stage analysis, even though more powerful tools, such as CFD calculations, are now taking over.

The fluid is split up into a fixed mass, which does mot move and is static relative to the tank, and a number of slosh masses connected to pendulums and relative dashpots. These slosh masses are given by the number of slosh modes considered in the analysis. For axi-symmetric tanks it has been shown [14] that the first mode is sufficient to provide a proper mechanical analogy, therefore in this paper no more than one mode is considered.

The mechanical slosh parameter models cited in the section are valid if slosh displacements are small. If the wave amplitude is greater than 10% of the tank diameter, splashing, breaking waves and even rotary sloshing will occur, thus the apparent slosh damping will increase in the mechanical slosh model [14].
In this regard, three different dynamic regimes for the motion of the liquid free surface can be identified, each requiring a different model [15]. More specifically, they are:

- linear motion: it corresponds to the situation in which limited accelerations and correspondingly limited liquid oscillations are considered. The liquid free surface happens to remain approximately planar.

- weakly non-linear motion: when oscillations have a frequency closer to the natural frequencies of the system. It is here that rotary sloshing shows up, causing the liquid to start swirling while oscillating.

- strongly non-linear motion: this describes the situation when the free surface is subjected to high accelerations and impacts against the container wall.

The more complex the motion of the liquid, the more complicated the equivalent model. In many practical cases, though, linear or at most weakly non-linear models are reasonably accurate and therefore in this paper attention is brought only to the linear, i.e. the simplest, type of motion.

2.1.1 Mechanical model parameters

The parameters of interest are the slosh wave frequency (f_S), slosh mass (m_S), pendulum hinge point location (H_S) and finally the damping ratio (ζ). All of these are a function of the time-dependent filling grade of the tank (h) and the tank radius (R).

For the purposes of the analysis the fixed mass is not considered, even though after the tank has significantly emptied all the remaining liquid can slosh and the fixed mass becomes zero. It has been shown that the fixed mass becomes zero when the filling grade, $x_{filling} = h'/h \simeq 33\%$, where h' and h are the current and maximum filling levels respectively [2]. In general, the fixed mass (m_F) can be calculated as follows:

$$m_F = m_{liquid} - \sum_{j=1}^{n_s} m_{S,j} \tag{2.1}$$

where $m_{S,j}$ is the slosh mass of the $j-th$ mode considered in the analysis and m_{liquid} is the overall mass of propellant stored in the tank at that time.
The latter can be calculated as: $m_{liquid} = \pi R^2 h \rho_{liq}$.

The scheme in figure 2.1 is a visual representation of the mechanical model used in the paper, for better understanding all the relationships between said parameters.

The slosh frequency can be calculated as follows:

$$f_s = \frac{1}{2\pi}\sqrt{1.841\frac{g}{R}\tanh\left(1.841\frac{h}{R}\right)} \tag{2.2}$$

where $g = 9.81 m^2/s$ is the gravitational acceleration.

The slosh mass is:

$$m_s = m\frac{R}{2.199h}\tanh\left(1.841\frac{h}{R}\right) \tag{2.3}$$

The pendulum hinge point location is:

$$H_s = h - \frac{R}{1.841}\left(\tanh\frac{0.9205h}{R} - \operatorname{csch}\frac{1.841h}{R}\right) \tag{2.4}$$

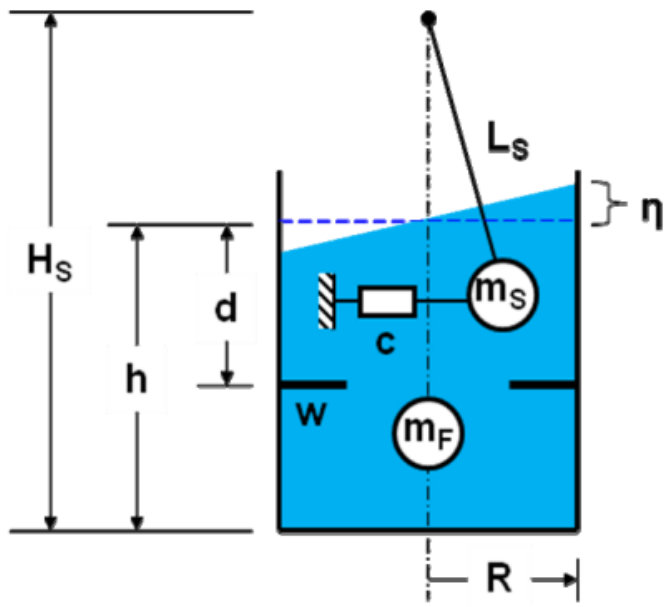


Figure 2.1: Parameters of mass-pendulum-dashpot mechanical model

The damping ratio can be evaluated from the damping factor. The latter is obtained from the logarithmic decrement of the slosh force decay curve.
The Force Decay Method consists of driving the tank to a frequency as close as possible to the resonant frequency of the principal slosh mode. Then the motion is abruptly stopped when the wave height tops a given value and the force in the drive link, which the tank is connected to, is recorded. The peak wave amplitudes are assumed to be proportional to the peak force amplitudes, thus enabling the use of the ratio between successive peak forces to compute the damping factor [16].

More precisely, the amplitude of the slosh force at a given cycle is represented by the letter x and the number of cycles over which the decay is measured is represented by the letter n. Therefore, x_0 is the amplitude at the initial moment and x_n at the $n-th$ cycle. More in detail the damping factor, δ:

$$\delta = \frac{1}{n} ln \left(\frac{x_0}{x_n} \right) \tag{2.5}$$

Accordingly, the damping ratio, ζ, can be defined in terms of the damping factor, δ, as follows:

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}} \tag{2.6}$$

These expressions are adopted to extract the damping of a system from experimental results, where the consecutive peaks in the free decay curve are used to calculate the damping factor and consequently the damping ratio.

When estimating the effectiveness of anti-sloshing devices such as inner mounted baffles, the damping ratio is the only parameter needed to compare the results between different configurations, including the bare-wall tank. In such a case the damping mechanism is offered by the interaction between the fluid and the tank wall and a general equation of smooth-wall damping is adopted.

2.1.2 Smooth-wall damping

The smooth-wall damping is the damping produced by the friction forces, due to the *shear stresses*, occurring between the tank wall and the fluid. It does not provide such a meaningful damping effect, as it shows the need for other specific devices, such as baffles, to accomplish this action. The magnitude of smooth-wall damping has been determined for several configurations and it can be described by a semi-empirical equation proposed by Mikishev and Dorozhkin [17], of the form:

$$\zeta_{SW} = \frac{4.98}{2\pi}\sqrt{\frac{\nu}{R^{1.5}g^{0.5}}}\left[1+\frac{0.318}{\sinh(1.84h/R)}\left(1+\frac{1-(h/R)}{\cosh(1.84h/R)}\right)\right]K \tag{2.7}$$

Where ζ_{SW} is the smooth-wall damping, namely the "viscous damping coefficient", ν is the kinematic viscosity of the liquid and K is a numerical coefficient that takes on different values depending upon tank geometry and fill level. For a cylindrical tank, $K = 1$.
More detailed expressions for other tank geometries are collected in the Abramson's textbook [7].

In the tank model de-ionized water was used as fluid medium throughout all tests, with a kinematic viscosity of $\nu = 0.001566 in.^2/s$, whereas the standard unit for ν is m^2/s. To compensate this, a conversion factor from $in.^2/s$ to m^2/s is required in the formulation for smooth-wall damping, which holds the following value: 0.00064516. In the analytical formulation for the smooth-wall damping this factor has been rounded off to 0.001 as it shows very good agreement with experimental data.
Eventually, the magnitude of smooth-wall damping will be even smaller for the full-scale tank due to its dimensions and the propellant properties.

In addition to that, since the damping is a function of the fluid viscosity and tank radius, the damping value for bare-wall tanks must be scaled according to the Reynolds number. The dimensionless version of the above equation will have to take into consideration this dependency to provide a general, fluid-independent, damping expression.

2.1.3 Baffle damping

Due to the inefficacy of smooth-wall damping, other mechanisms for suppressing the sloshing inside a container are required, thus explaining the development of such a wide array of anti-slosh systems as listed in the previous chapter 1.

The damping equations for flat ring baffles have already been discussed in the dedicated section 1.3. Since this design corresponds to the tested devices, no further investigation

into baffle expressions for different geometries is suggested.

Unlike the smooth-wall case, slosh damping is here nearly independent of the working fluid, therefore no correction factors need to be accounted for. In addition to that, baffle damping scales one-to-one in terms of logarithmic decrements for geometrically scaled baffles and slosh wave amplitudes [4].

To this extent, the reader is recommended to consult the already presented Miles equation in its simplified form (1.8) for a better understanding.

Baffle damping has been experimentally determined with baffles installed in a variety of configurations in the tank model and with varying fluid levels. In particular, this paper focuses on testing a single-baffle configuration and comparing the resultant damping with a four-baffle configuration. In the latter situation, the damping of a series of baffles is calculated on the basis of the linear superposition principle, therefore no complex equations are involved.

2.2 Experimental data and analysis results

2.2.1 Model scaling

This section constitutes the core of the thesis as the aforementioned mechanical model parameters are validated using experimental results and the accuracy of the analytical formulations is here tested. The primary advantage of testing and extrapolating the parameters of the equivalent model is that the force and damping response on sub-scale tanks can be used to determine these parameters for the full-scale versions, if adequately non-dimensionalized.
On top of that, if a good agreement between test data and analysis is attained, the validated mathematical expressions are suitable to describe other designs involving different tank sizes as well as other propellant properties. By doing so, the reliability of such equations would be enough to estimate the mechanical model parameters notably well in the preliminary design of a complex system.

The experiments here shown were performed on a sub-scale tank model of the Ares I US LH2 tank, with a 1/5 scaling ratio. This resulted in a 43 inches (110 cm) wide tank model and a height of the barrel section of approximately 45 inches (115 cm). A schematic of the tank model is shown in figure 2.2. Test data are obtained from a paper by *Perez et al., 2012* [4].
The baffles installed have different width between the single-baffle configuration and the four-baffle one, but they are all made of 0.12 inch thick stainless steel sheets.

Despite all these tests having provided results depending on the tank geometry and fluid properties (except for the baffle damping), non-dimensionalized parameters are presented in this paper. This accounts for more general and especially geometry-unrelated test results, allowing comparisons to be drawn more easily between different configurations, such as cylindrical tanks rather than spherical tanks and so on.

To this extent, a comparison table between original and dimensionless parameters is below presented:

Parameter	Original	Non-Dimensionalized
Fill Level	h	h/R
Slosh Frequency	f_S	$\omega_S R^{0.5}/g^{0.5}$
Slosh Mass	m_S	m_S/m
Pendulum Hinge Point	H_S	H_S/R
Viscous Damping	ζ_{SW}	$\zeta_{SW} R^{0.75} g^{0.25}/\nu^{0.5}$
Drag Damping	ζ_{Miles}	ζ_{Miles}

Table 2.1: Comparison table between original and dimensionless parameters

In the table, $\omega_S = 2\pi f_S$ is the natural frequency of the oscillating pendulum and is measured in rad/s.

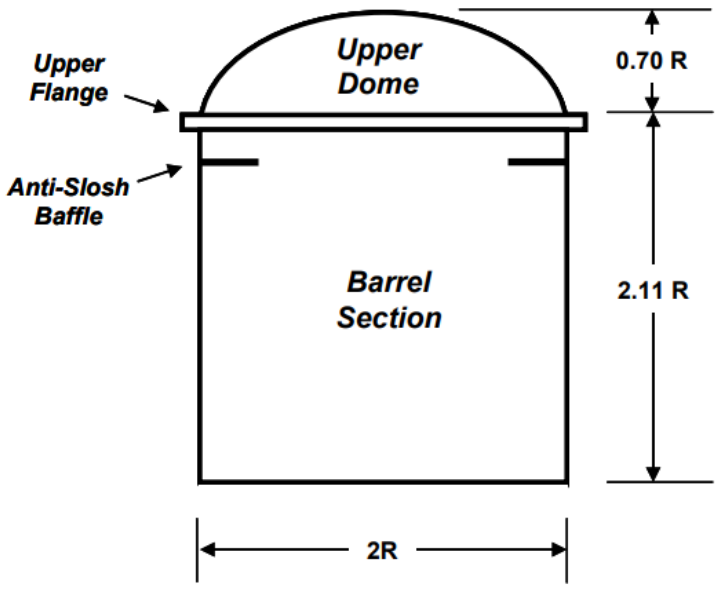


Figure 2.2: Ares I US LH2 tank model design

2.2.2 Test results of slosh model parameters

The analysis and test results of the slosh mechanical model based on the Ares I US LH2 tank model are reported in the following tables and they are graphically represented in the following figures. The tables collect data about the non-dimensionalized slosh frequency, slosh mass, pendulum hinge point location and the damping offered by smooth wall and two baffle configurations.

Each table has one column for the fill levels at which the measurement was done, one column for the corresponding tested value, one for the analytical solution at that fill level and one final comparative column. The latter shows the discrepancy between analysis and testing through a percent difference with respect to analysis.
The outcomes of this comparison are briefly discussed after each representation.

Slosh frequency

Table 2.2: Ares I LH2 tank model - Slosh frequency Test 1

Fill level h/R	**Test 1 slosh frequency** $\omega_S(R/g)^{0.5}$	**Analysis slosh frequency** $\omega_S(R/g)^{0.5}$	**Difference** %
0.192	0.782	0.791	1.08
0.321	0.986	0.988	0.24
0.385	1.06	1.06	0.03
0.577	1.20	1.20	0.28
0.661	1.24	1.24	0.21
0.745	1.27	1.27	0.17
0.977	1.32	1.32	0.01
1.21	1.34	1.34	0.09
1.39	1.35	1.35	0.09
1.6	1.35	1.35	0.23
1.82	1.35	1.36	0.38
2.08	1.35	1.36	0.46
2.17	1.35	1.36	0.47
2.28	1.40	1.36	3.20

Table 2.3: Ares I LH2 tank model - Slosh frequency Test 2

Fill level h/R	**Test 2 slosh frequency** $\omega_S(R/g)^{0.5}$	**Analysis slosh frequency** $\omega_S(R/g)^{0.5}$	**Difference** %
1.3	1.33	1.35	1.16
1.9	1.34	1.36	1.15
2.3	1.41	1.36	3.94

Slosh frequency has been observed to increase with fill level, up to a certain point where it becomes nearly constant. The analytical expression is particularly consistent with the experimental results, leading to a maximum 3.2% difference.

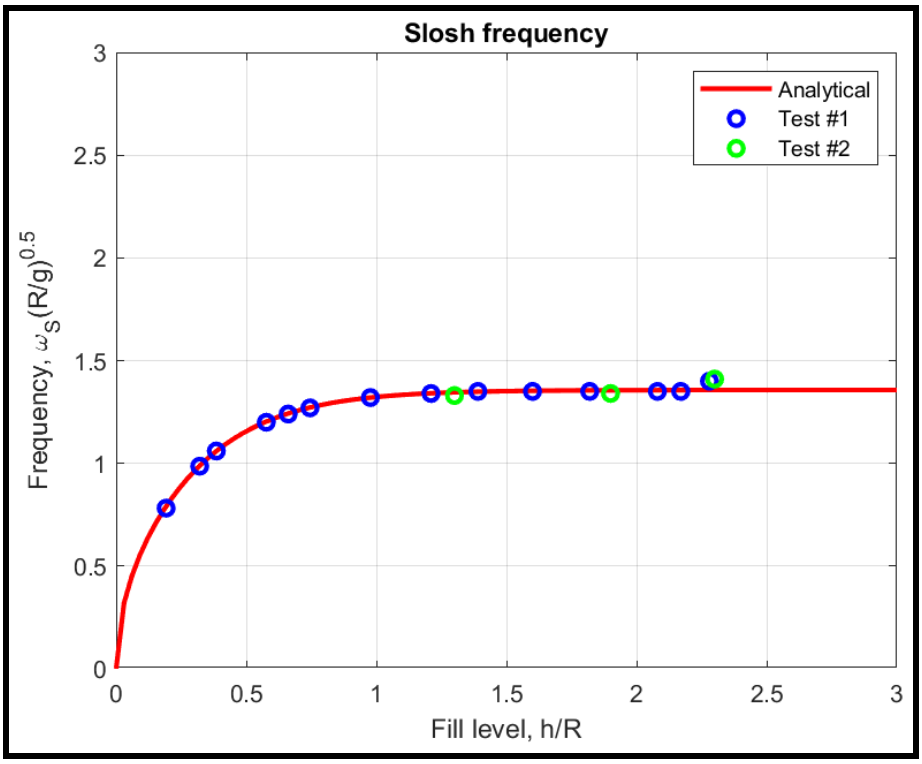


Figure 2.3: Ares I LH2 Tank Model - Slosh frequency

Slosh mass

Slosh mass equation also shows an acceptable agreement with experimental data, as it decreases with increasing fill levels. This suggests that when the tank is almost full the sloshing mass, i.e. the damping requirement for stable attitude control, is very little compared to when the tank is nearly empty. In fact, as it has already been discussed, when the fill level drops below approximately 33% of the total capacity, the fixed mass in the mechanical model becomes zero and all the liquid in the tank starts moving.

Table 2.4: Ares I LH2 tank model - Slosh mass Test 2

Fill level h/R	**Test 2 slosh mass** m_S/m	**Analysis slosh mass** m_S/m	**Difference** %
1.3	0.317	0.344	7.85
1.9	0.231	0.239	3.31
2.3	0.19	0.198	3.86

Table 2.5: Ares I LH2 tank model - Slosh mass Test 1

Fill level h/R	**Test 1 slosh mass** m_S/m	**Analysis slosh mass** m_S/m	**Difference** %
0.192	0.801	0.804	0.37
0.321	0.764	0.752	1.64
0.385	0.75	0.72	4.11
0.577	0.599	0.62	3.37
0.661	0.558	0.577	3.30
0.745	0.556	0.537	3.62
0.977	0.448	0.441	1.67
1.21	0.37	0.367	0.76
1.39	0.338	0.323	4.56
1.6	0.29	0.283	2.60
1.82	0.245	0.249	1.71
2.08	0.23	0.218	5.30
2.17	0.212	0.209	1.23
2.28	0.19	0.199	4.70
2.44	0.149	0.186	20.03

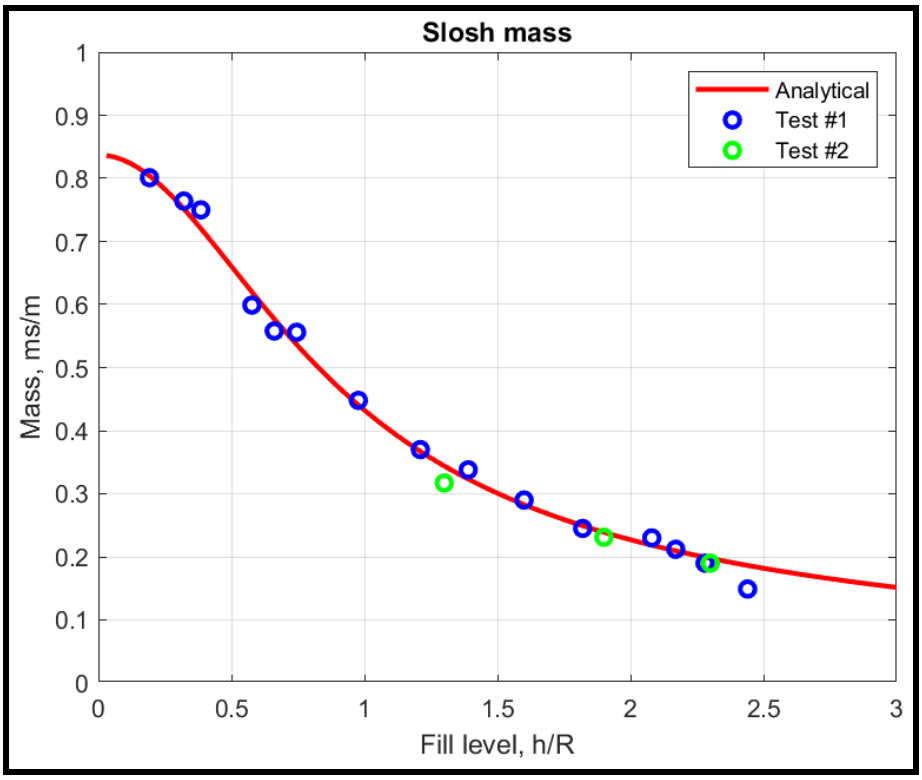


Figure 2.4: Ares I LH2 Tank Model - Slosh mass

Hinge point location

Table 2.6: Ares I LH2 tank model - Hinge point location Test 1

Fill level h/R	**Test 1 hinge location** H_S/R	**Analysis hinge point** H_S/R	**Difference** %
0.192	1.52	1.51	0.48
0.321	1.02	0.901	13.26
0.385	0.931	0.759	22.59
0.577	0.744	0.576	29.11
0.661	0.74	0.558	32.6
0.745	0.789	0.562	40.36
0.977	0.847	0.648	30.78
1.21	0.96	0.798	20.34
1.39	1.09	0.938	16.22
1.6	1.28	1.12	14.58
1.82	1.43	1.32	8.64
2.08	1.68	1.56	7.63
2.17	1.78	1.65	8.06
2.28	1.76	1.75	0.38
2.44	1.80	1.91	5.72

Table 2.7: Ares I LH2 tank model - Hinge point location Test 2

Fill level h/R	**Test 2 hinge location** H_S/R	**Analysis hinge point** H_S/R	**Difference** %
1.3	1.03	0.866	18.96
1.9	1.55	1.39	11.45
2.3	1.87	1.77	5.48

Hinge point location curve shows bigger inconsistency with test data, as the maximum difference amounts to 40% with respect to the analytical formulation. This could be explained by the fact that coefficients in the formula are not appropriate for this tank configuration, as the curve appears to have a very similar trend with acquired data. It seems as if the curve is only translated to lower y values, still describing the accurate evolution of test records.

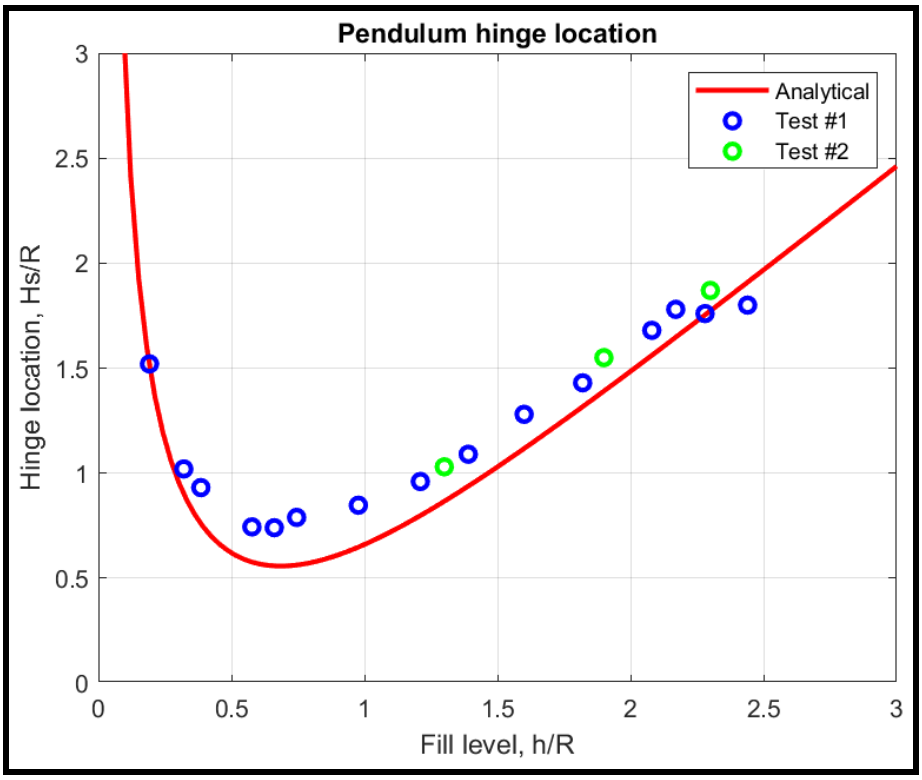


Figure 2.5: Ares I LH2 Tank Model - Hinge point location

Smooth wall damping

For the evaluation of smooth-wall damping two tests were performed on the same bare tank with two different slosh wave amplitudes. The first test was performed with a nominal slosh wave amplitude of 1 inch ($\eta/R = 0.0458$) and the second one of 2 inches ($\eta/R = 0.0916$). As figure 2.6 illustrates, no significant difference is observed for the two slosh wave amplitudes tested. As expected, smooth-wall damping is independent of slosh wave amplitude for $\eta/R \leq 0.1$ [14].
Test damping in the table is the average value of the damping extracted from the two measurements.

In smooth wall damping as fill level decreases, the wetted area to the slosh volume increases, therefore driving an increase in the damping value. Overall, test results are consistent with analytical values and they show an increase in damping in the upper dome region (for fill levels $h/R \geq 2.11$, see figure 2.2). There is a slight discontinuity at or just above this liquid level because of a misaligned aluminum ring placed between the upper flanges of the tank to prevent water leaking during testing.
The percent difference increases for higher fill levels.

Table 2.8: Ares I LH2 tank model - Smooth-wall damping

Fill level h/R	**Test damping (%)** ζ_{SW}	**Analysis damping (%)** ζ	**Difference** %
0.192	0.254	0.202	25.58
0.321	0.162	0.143	13.49
0.385	0.137	0.128	7.03
0.577	0.104	0.104	0.23
0.661	0.101	0.0987	2.37
0.745	0.0891	0.0946	5.82
0.977	0.088	0.0879	0.10
1.21	0.0742	0.0845	12.21
1.39	0.07295	0.083	12.06
1.6	0.0762	0.0818	6.79
1.82	0.08045	0.0809	0.59
2.08	0.09735	0.0803	21.22
2.17	0.10435	0.0801	30.19
2.23	0.1184	0.0801	47.89
2.28	0.0982	0.08	22.77
2.44	0.122	0.0798	52.87

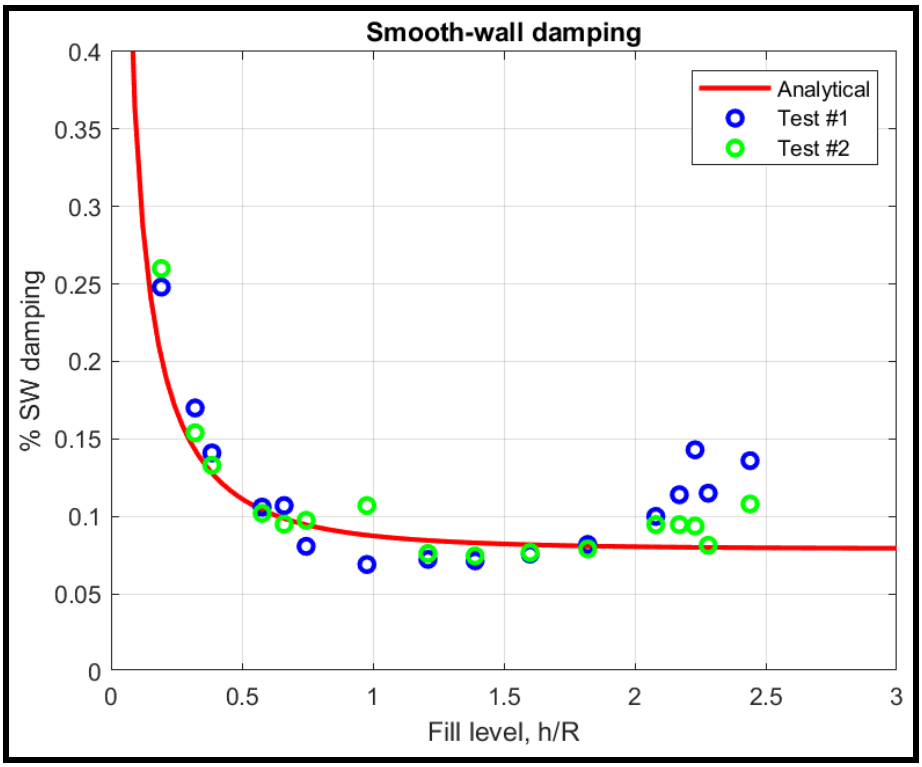


Figure 2.6: Ares I LH2 Tank Model - Smooth wall damping

Single baffle damping

The baffle tested was 4.4 inches wide ($w/R = 0.204$) and was located at height of 43.91 inches ($h/R = 2.04$). Since the baffle damping is dependent on the slosh wave amplitude, only the first two peaks of the force decrement data were considered to obtain the logarithmic decrement. The average value of all slosh wave amplitudes measured during testing of this series is $\eta/R = 0.0585$ and it was inserted into the Miles equation to create the analytical curve shown in figure 2.7.

Table 2.9: Ares I LH2 tank model - Single baffle damping

Fill level h/R	**Test damping (%)** ζ	**Analysis damping (%)** ζ	**Difference** %
1.93	0.11	0.0806	36.43
2.08	19.2	12.60	52.04
2.09	20.8	12.10	72.47
2.11	11	11	0.00
2.16	7.95	8.74	9.03
2.21	5.62	6.94	19.06
2.23	5.23	6.33	17.42
2.26	3.74	5.52	32.20
2.28	3.68	5.03	26.86
2.36	2	3.48	42.56
2.47	1.04	2.1	50.46

Baffle damping reaches a peak value when the device is located just below the liquid free surface, decreasing sharply for higher baffle depths. For lower fill levels, where the baffle is above the liquid free surface the smooth wall damping still takes place, making for a significantly weaker damping effect.

For this parameter, the two analysis expressions discussed in the ring baffle section 1.3 have been plotted to observe how one differs from the other. In particular, the *Miles#1* curve refers to the more general equation 1.8, whereas *Miles#2* is the simplified one for cylindrical tanks 1.11. The second one is slightly more accurate compared to the first one.

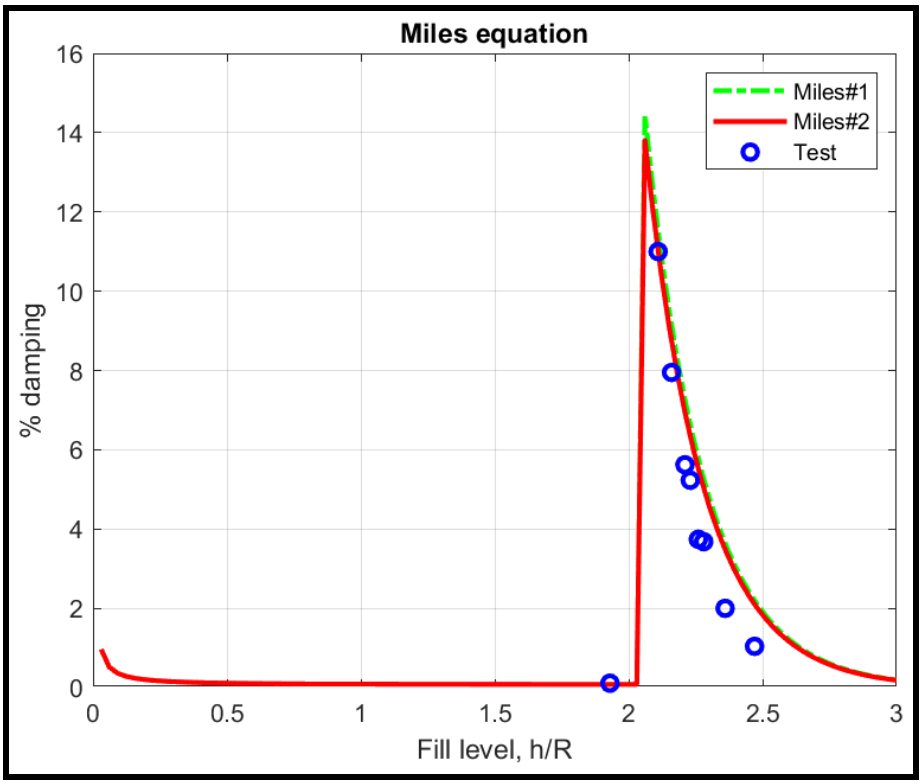


Figure 2.7: Ares I LH2 Tank Model - Miles equation, single baffle

However, some divergence is noticed for increasing fill levels, again questioning the accuracy of some coefficients in the formula. Another reason for the discrepancy could be attributed to the fact that experimental data have been obtained through a logarithmic decrement and some uncertainties in the measurement are inevitably present. The maximum difference for baffle damping is recorded to be 50% at a fill level $h/R = 2.47$.

Since the trend is very similar to real damping, a new coefficient for the exponent is proposed, as it shows more compliance with experimental results. The corrected version of the Miles equation would become:

$$\zeta_{Miles} = 2.83e^{-6d/R}\alpha^{1.5}\sqrt{\frac{\eta}{R}} \tag{2.8}$$

Where the previous *-4.601* coefficient in the exponent is replaced with a higher absolute value, roughly *-6* in the new version.

The following table and figure represent how this version of the Miles equation better fits experimental data, resulting in a more accurate approximation.

Table 2.10: Single baffle damping - Miles equation corrected

Fill level h/R	**Test damping (%)** ζ	**Analysis damping (%)** ζ	**Difference** %
1.93	0.11	0.0806	36.43
2.08	19.2	11.9	60.79
2.09	20.8	11.2	84.96
2.11	11	9.97	10.29
2.16	7.95	7.39	7.59
2.21	5.62	5.47	2.67
2.23	5.23	4.85	7.73
2.26	3.74	4.06	7.77
2.28	3.68	3.6	2.32
2.36	2	2.23	10.13
2.47	1.04	1.15	9.58

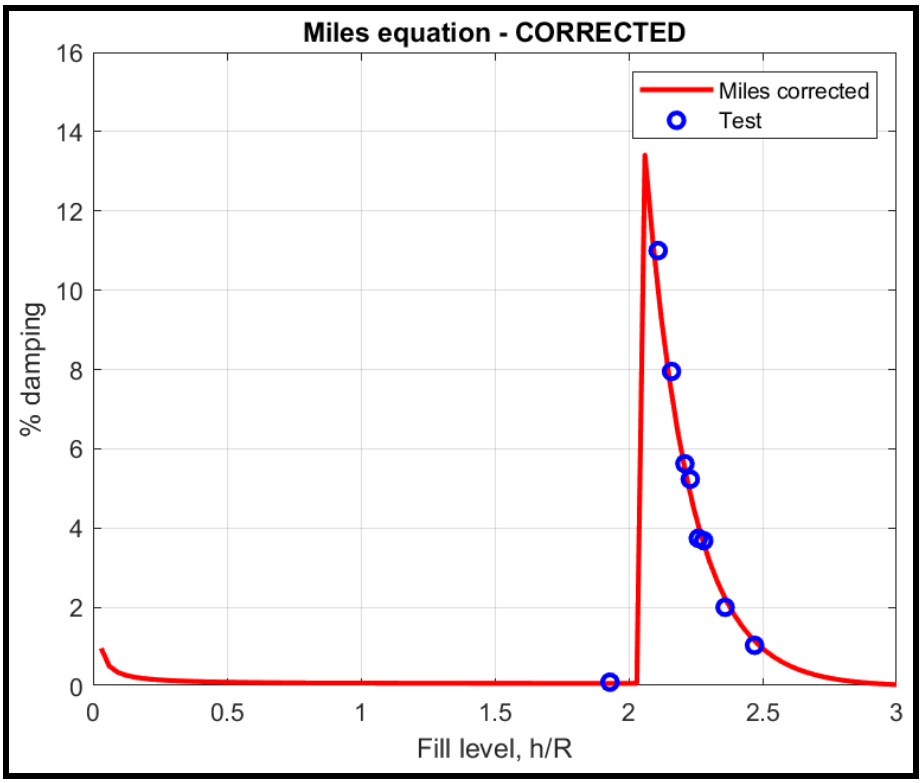


Figure 2.8: Miles equation with different exponent

With a higher value in the exponent, the maximum percent difference sticks around 10% for a fill level of $h/R = 2.36$ when considering baffle damping. The spike in percent

difference (nearly 85%) observed in table 2.10 shows that the analytical formula fails to describe the fluid behavior in the close proximity of the inner mounted baffle, due to some instationary flow and complicated fluid dynamics around sharp edges. It only predicts accurate values for greater baffle depths.

Multiple baffles damping

In this final section a configuration involving four flat ring baffles is investigated. Baffles here installed are reasonably smaller compared to the single baffle design above discussed, as they have a relative width of $w/R = 0.0533$.

These baffles locations are: $h/R = 2.04, 1.77, 1.50, 1.24$ for the $\#1, \#2, \#3, \#4$ baffle respectively.
A nominal slosh wave amplitude of 1 inch ($\eta/R = 0.0464$) was used in the test.

Table 2.11: Ares I LH2 tank model - Multiple baffle damping

Fill level h/R	**Test damping (%)** ζ	**Analysis damping (%)** ζ	**Difference** %
1.3	1.76	1.55	13.85
1.37	1.09	1.12	2.70
1.44	0.904	0.812	11.36
1.57	2.42	1.92	25.86
1.64	1.52	1.39	9.09
1.7	0.969	1.06	8.34
1.84	2.09	2.03	2.88
1.9	1.53	1.54	0.75
1.97	1.02	1.12	8.69
2.1	2.31	2.16	6.94
2.17	1.41	1.57	9.92
2.24	1.19	1.13	4.91
2.3	0.639	0.861	25.76

In this multiple-baffle configuration, an excellent agreement between test data and analysis is recorded. This validates the assumption that in cylindrical tanks, for a given fill level, the damping contribution of multiple baffles can be added in series (linear superposition). The % damping is smaller with respect to the previous design, figure 2.7, because of the reduced baffle width.

Above all, both experimental and analytical results demonstrate the reason why a multiple-baffle design is sometimes preferred. More precisely, if more than a single baffle is used, the damping provided by these devices is more uniform throughout all fill levels. Baffles located higher in the tank compensate for the insufficient damping provided by baffles at lower positions (for high filling grades), as the latter start working properly only as fill level decreases.

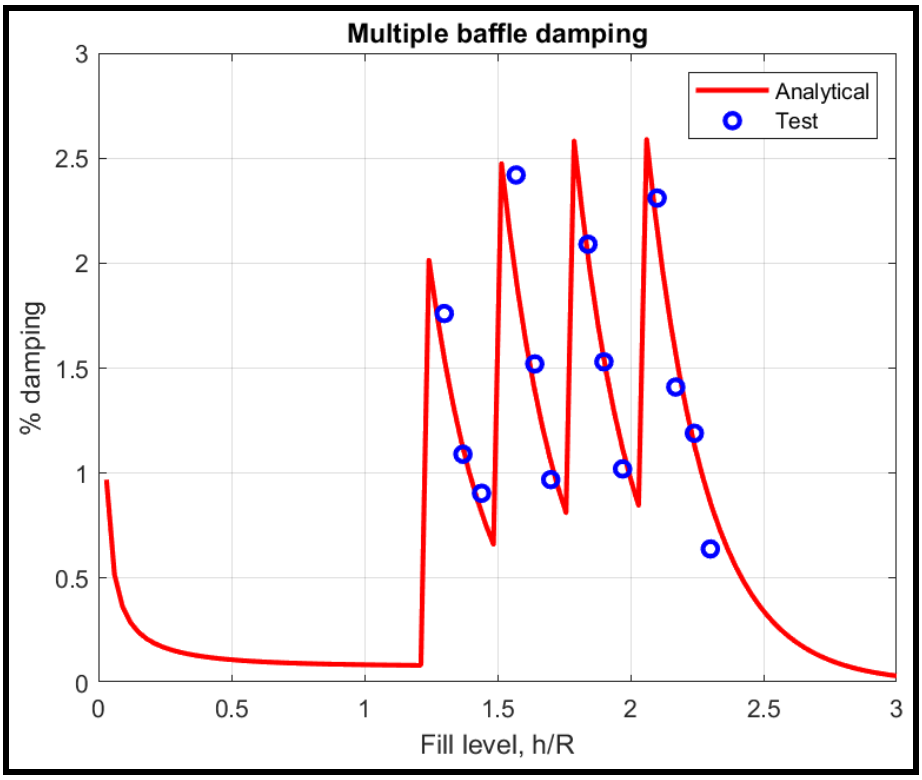


Figure 2.9: Ares I LH2 Tank Model - Miles equation, multiple baffles

2.3 Damping comparison

2.3.1 Equation of motion

This section targets the comparison in the damping effect between a bare tank, where only smooth-wall damping is present, and a single-baffle configuration, with Miles damping expression.

This comparison will be evaluated through a free attenuation curve of the slosh mass displacement with respect to the equilibrium point, i.e. when the slosh mass is at rest ($\theta_{eq} = 0$). This corresponds to a single degree of freedom system, where the equation of motion in its critical damping form is:

$$\ddot{\theta} + 2\zeta_n\omega_n\dot{\theta} + \omega_n^2\theta = 0 \tag{2.9}$$

Where ω_n is the natural frequency of the $n-th$ sloshing mass, which is calculated from equation 2.2 and ζ_n is the respective damping ratio, which can either be obtained from the smooth-wall damping (2.7) for the bare tank or from Miles equation (1.11) when a baffle is inserted. For the purposes of this investigation only the first slosh mode is considered.

Since the mass-pendulum-dashpot model is a linear discrete model, some important assumptions are made. Firstly, the vertical displacement of the liquid center of mass is neglected and secondly the liquid free surface is considered to remain planar throughout the oscillation. This allows us to find a linear relationship between the sloshing wave height at the wall of the container and the pendulum swinging angle:

$$\bar{\eta} = Rtan(\theta) \tag{2.10}$$

Where $\bar{\eta}$ is the total sloshing height at the container wall and θ is the angular displacement of the slosh mass. However, it is important to bear in mind that the larger the angle (θ), the less reasonable is the initial assumption of a planar liquid free surface, and thus the less accurate are the expected results.

For a simple pendulum the natural frequency can be evaluated as a function of the pendulum length:

$$\omega_0 = \sqrt{\frac{g}{L_S}} \tag{2.11}$$

Being ω_0 the natural frequency, L_S the pendulum length, as shown in figure 2.1, and $g = 9.81 m/s^2$ the gravitational acceleration.

By plugging the above equation into the slosh frequency equation 2.2 it is possible to obtain the value of L_S:

$$L_S = \frac{R}{1.841 \tanh\left(1.841 \frac{h}{R}\right)} \tag{2.12}$$

This expression turns out to be useful for estimating the sloshing wave height, providing a more useful relationship with respect to equation 2.10. Indeed, with the approximation of small displacements, $\tan(\theta) \approx \sin(\theta) = x/L_S$, being x the horizontal displacement of the slosh mass [15]. Therefore, the sloshing height now becomes:

$$\bar{\eta} = 1.841 \tanh\left(1.841 \frac{h}{R}\right) x \tag{2.13}$$

This expression is very powerful as it allows to calculate the slosh mass displacement given the experimentally determined slosh wave height at the container wall, just by reversing the equation:

$$x = \frac{\bar{\eta}}{1.841 \tanh\left(1.841 \frac{h}{R}\right)} \tag{2.14}$$

Which gives the initial condition of the free attenuation curve, namely the initial (and maximum) mass displacement when the external force is suddenly stopped, as described in the subsection 2.1.1.

2.3.2 Slosh mass displacement

To find a solution to equation 2.9 it is first necessary to distinguish between an overdamped system ($\zeta > 1$), a critically damped system ($\zeta = 1$) or an underdamped system ($\zeta < 1$). Since the damping ratios involved in the analysis are all well below the unit value, the system is considered to be underdamped.

The general solution of the displacement as a function of time for an underdamped system is the following:

$$x(t) = C \sin(\omega_d t + \phi) e^{-\zeta \omega_0 t} \tag{2.15}$$

Where the constant C and the phase angle ϕ depend on the initial conditions and ω_d is the frequency of the damped oscillations, $\omega_d = \omega_0 \sqrt{1 - \zeta^2}$.
We now use the displacement variable x because it is more practical and can be scaled back to θ anytime through just the pendulum length, L_S.

The parameters used to determine these coefficients are obtained from the test results and they have been dimensionalized by multiplying the radius of the tank model. In particular, the smooth-wall damping and single-baffle damping results are compared for a fixed fill level of $h/R = 2.16$, which is slightly above the baffle location ($h/R = 2.04$). As far as the nominal slosh wave is concerned, the value given by the reference paper [4] for the single-baffle damping was utilized. In particular, at the said fill level, a slosh wave height of $\eta/R = 0.0669$ was reported. Therefore, this same value was also chosen for the smooth-wall damping.

After having dimensionalized all the variables and having set the phase angle $\phi = 0$, the value for the coefficient becomes $C = x_0 = 2.01$ cm where x_0 is the initial (and maximum) mass displacement, for $t = 0$ s. This value is calculated by using equation 2.14.

It is now possible to plot the equation of motion and compare the free attenuation curve of the two damping mechanisms. For a better visualization, two different time series are presented, one is over a short time interval (figure 2.10) and the other one covers a longer time span (figure 2.11).

In the figure, the lines in red and green represent the oscillation of the sloshing mass as a function of time with respect to the equilibrium position ($x = 0$). The blue and cyan lines, instead, are the so-called *extinction curves* as they delimit the peaks in the graph of $x(t)$. The expression for a generic extinction curve is as follows:

$$y(t) = \pm C e^{-\zeta \omega_0 t} \tag{2.16}$$

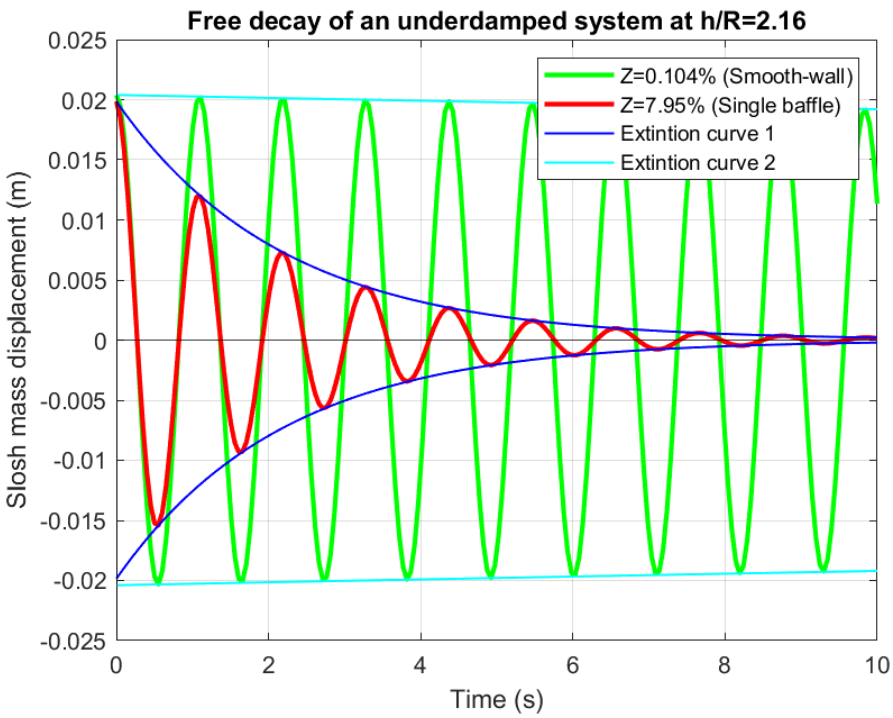


Figure 2.10: Slosh mass displacement with different damping ratios - smaller time interval

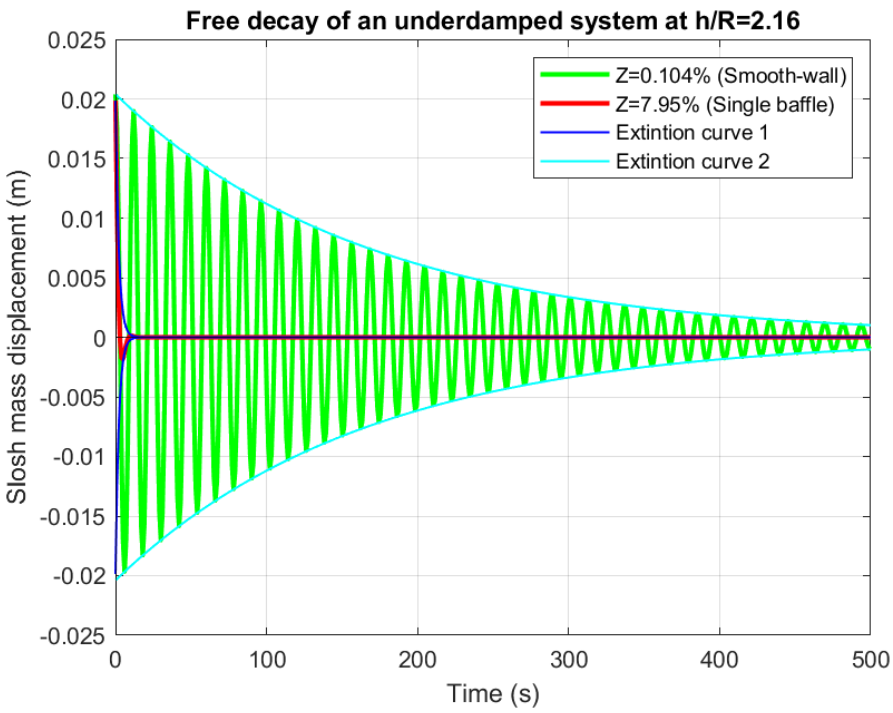


Figure 2.11: Slosh mass displacement with different damping ratios - longer time interval

2.3.3 Results comparison

These curves show a remarkably different result in terms of damping and the time it takes to the oscillation before coming to rest. In particular, it is clear that the single baffle performs an excellent job at extinguishing the oscillations, therefore proving its undeniable utility.

By solving a linear equation, it is possible to estimate how long it takes to the oscillating mass to stop the motion. The extinction curve is here set to equal 5% of the initial slosh mass displacement.

The time required to stop in the two setups is below presented.

	Maximum damping [%]	**Stopping time** [s]
Smooth-wall	0.104	~ 500
Single baffle	7.95	6.5

Table 2.12: Comparison table between smooth-wall damping and baffle damping

It is important to mention that the time solution found for the smooth-wall damping is a simple estimation and does not represent the real evolution. Indeed, the actual stopping time in the bare-wall tank will be smaller due to the viscosity of the fluid dissipating energy not only at the tank wall but also within the liquid itself. Moreover, the series of assumptions that have been made for this model are far from the real working conditions, sometimes resulting in imprecise estimations of the model parameters.

Nevertheless, the results give an idea of the difference in terms of orders of magnitude and validates the actual need for slosh-suppression devices, such as baffles, in tanks storing liquid propellants. The lower stopping time is beneficial with regard to attitude control, allowing the vehicle to follow more accurately the predetermined flight path.

Chapter 3

Implementation of the mechanical model

3.1 Efesto tank configuration

This chapter is dedicated to the implementation of the mechanical model discussed in chapter 2 as the validated formulas are now used to estimate the real damping and efficacy of baffles in a full-scale tank.

Since this thesis deals with a topic of interest for the Polito Rocket Team, the model tank will be the actual liquid oxygen tank used in the *Efesto* rocket. The LOX tank is made up of two hemispherical domes and a central barrel section. In particular, it features four baffles located in the cylindrical section, whose damping effect is here studied using the Miles equation. All the mechanical model parameters will be evaluated as well, after dimensionalization according to the tank geometry and liquid properties.

The geometry of the tank is displayed in figure 3.1, from which it is possible to observe that the baffles have been installed at the same non-dimensional locations as the test mock-up in section 2.2.2. In that figure the reported distances are measured from the beginning of the cylindrical section, as the lower hemispherical dome is not analyzed because it would require other mechanical model expressions. This is the reason why in this study the liquid fill level is assumed to always remain within the barrel section. Indeed, damping is here calculated, disregarding the top and bottom domes.

However, in the calculation of the total stored mass the contribution of the lower dome can not neglected.

The hemispherical dome radius is set to be $R = 95\,mm$, which is also the cylindrical tank radius. The height of the central section is 218,60 mm and defines both lower and upper limits of the liquid fill levels examined. Baffle width is set to be $w = 10\,mm$ with a baffle thickness of $t_b = 3.048\,mm$.

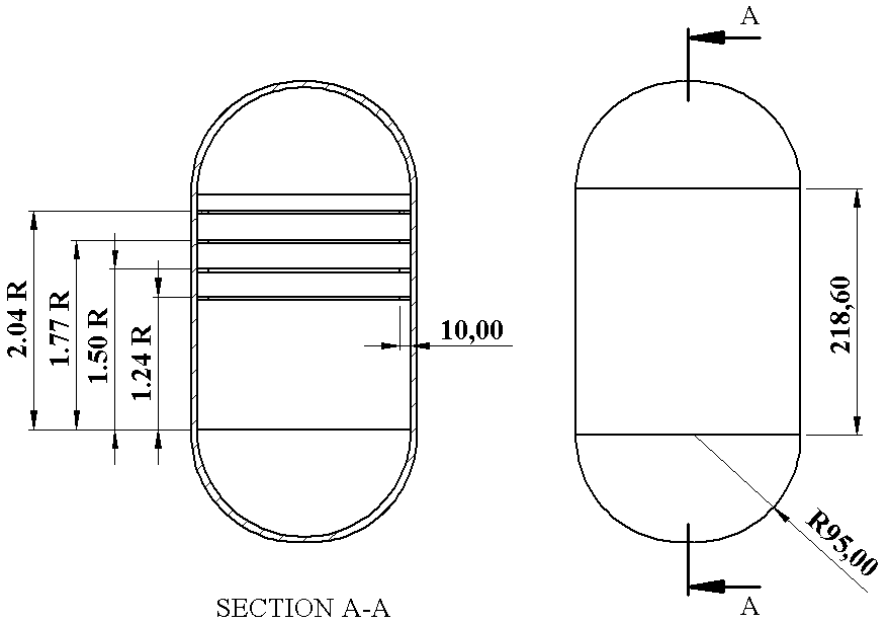


Figure 3.1: Efesto LOX tank geometry

The four-baffle design is summarized in table 3.1.

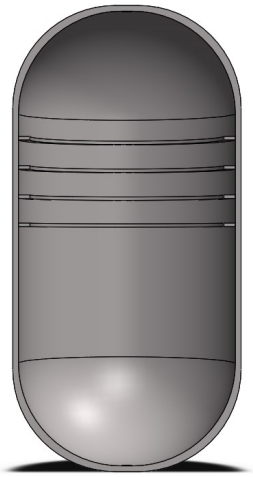
Baffle ID #	**Baffle width w/R**	**Baffle location h/R**
1	0.1053	2.04
2	0.1053	1.77
3	0.1053	1.50
4	0.1053	1.24

Table 3.1: Efesto LOX tank baffle design

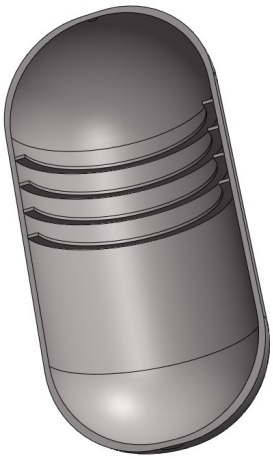
As far as the liquid properties are concerned, the values have been calculated using *Coolprop*, an external library in Python, which provides thermodynamic variables for any user-defined working environment. Having set the desired pressure and temperature constraints the value for the liquid oxygen density is approximately: $\rho = 1152\, kg/m^3$. Analogously, it is possible to calculate the dynamic viscosity of LOX under working conditions: $\mu = 2.043 \cdot 10^{-4}\, Pa\, s$. From these two values, the kinematic viscosity is derived as follows: $\nu = \mu/\rho = 1.774 \cdot 10^{-7}\, m^2/s$.
These values are needed to calculate dimensional parameters in the mechanical model.

For a better understanding of the configuration, a section view of the tank is below presented (figure 3.2):

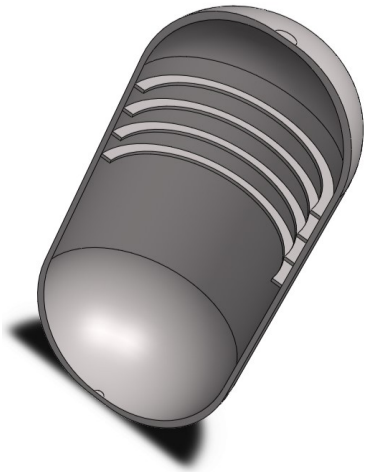
(a) Outside view of the tank



(b) Baffle layout inside the tank



(c) Lower-right view



(d) Upper-right view

Figure 3.2: Efesto LOX tank setup

3.2 Mechanical model

3.2.1 Pendulum model setup

All mechanical model parameters in this section are calculated using equations presented and thoroughly discussed in section 2.1. In particular, what differs from the previous investigation is that all the curves here illustrated are presented in a dimensional form such that the real values can be extracted from the model. The data is needed to later evaluate how the sloshing and the corresponding suppression mechanism are addressed.

For the slosh frequency, slosh mass and pendulum hinge point location equations 2.2, 2.3 and 2.4 have been used respectively.

In particular, the liquid mass used in the sloshing mass equation is calculated as a function of fill level starting from the cylindrical section. This is then added to the lower dome liquid mass, as follows:

$$m = \rho_{LOX} \left[\frac{2}{3}\pi R^3 + \pi R^2 \left(\frac{h}{R} - 1 \right) R \right] \tag{3.1}$$

Where h/R is the non-dimensional fill level measured from the bottom of the tank.

For the smooth-wall damping, equation 2.7 is used, while the multiple-baffle damping relies on the linear superposition of Miles equation 1.11.
The latter expression makes use of the slosh wave height and therefore a value needs to be provided for the calculation. This is found as a trade-off between having a relevant sloshing wave height at the tank wall (large η) and the limited accuracy of the linear pendulum model for large swinging angles (small η).

Therefore, a value of $\theta = 10°$ is chosen as maximum angular displacement. According to equation 2.10 the corresponding non-dimensional wave height is obtained: $\eta/R = 0.176$. This quantity is used to calculate both the damping offered by Miles equation and to evaluate the initial slosh mass displacement using equation 2.14.

It is important to observe that, compared to the previous Ares I US LH2 tank model, the configuration for the baffle setup is slightly different. Baffles in the Efesto tank have a bigger width-to-radius ratio, since the tank is substantially smaller. For the same reason, despite the non-dimensional slosh wave amplitude is larger, the actual wave height measured at the container wall turns out to be much smaller.

More precisely, for the Ares I US LH2 tank model the maximum recorded wave height in the multiple-baffle test was: $\eta_{max} = 0.0464\,R = 2.54\,cm$. On the other hand, in the Efesto LOX tank, as previously stated, it becomes:

$$\eta_{max} = \tan\left(\theta_{max}\right)\, R = 0.176\,R = 1.67\,cm \tag{3.2}$$

3.2.2 Dimensionalized parameters

All mechanical model parameters are represented as a function of fill level expressed in centimeters, ranging from the initial to the final sections of the cylindrical part, without evaluating any other liquid levels. The hemispherical domes are here neglected as they would require additional formulations due to their different shape, which goes beyond the purpose of this thesis.

Figures 3.3 to 3.7 graphically represent the trend of the dimensionalized parameters for the pendulum model.

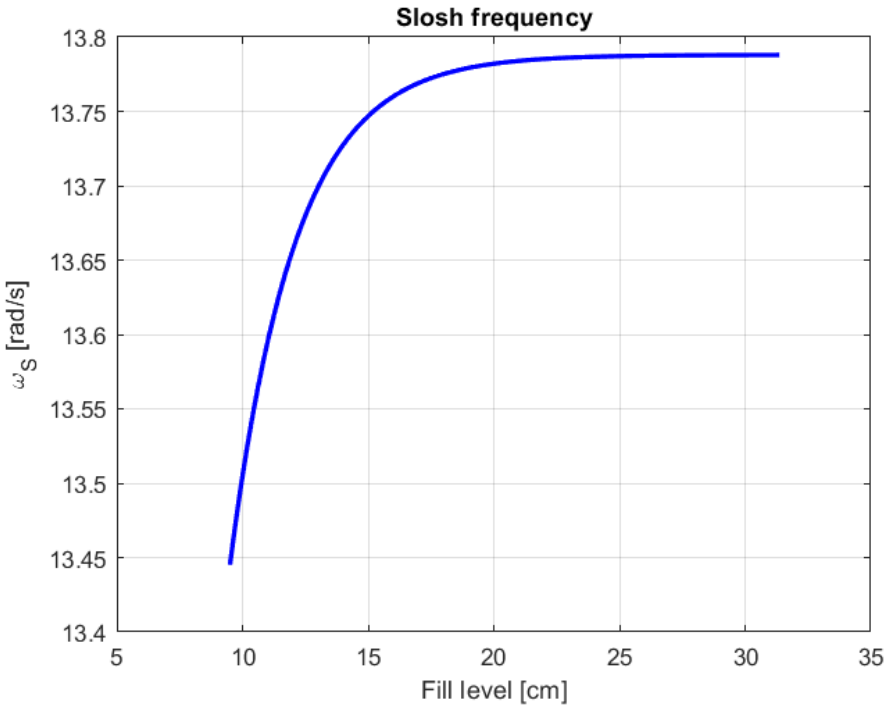


Figure 3.3: Efesto LOX tank - Slosh frequency

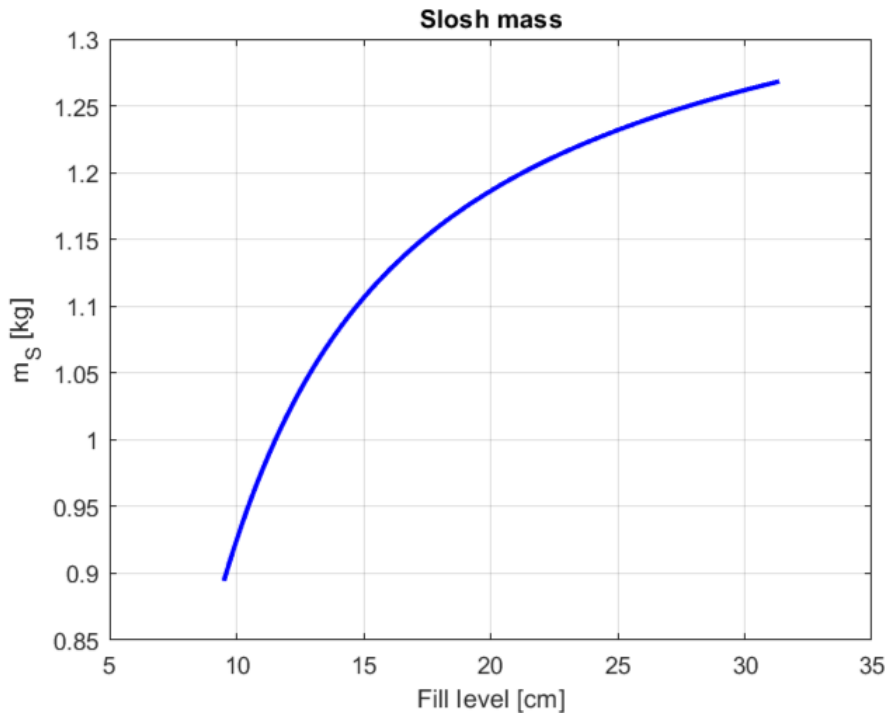


Figure 3.4: Efesto LOX tank - Slosh mass

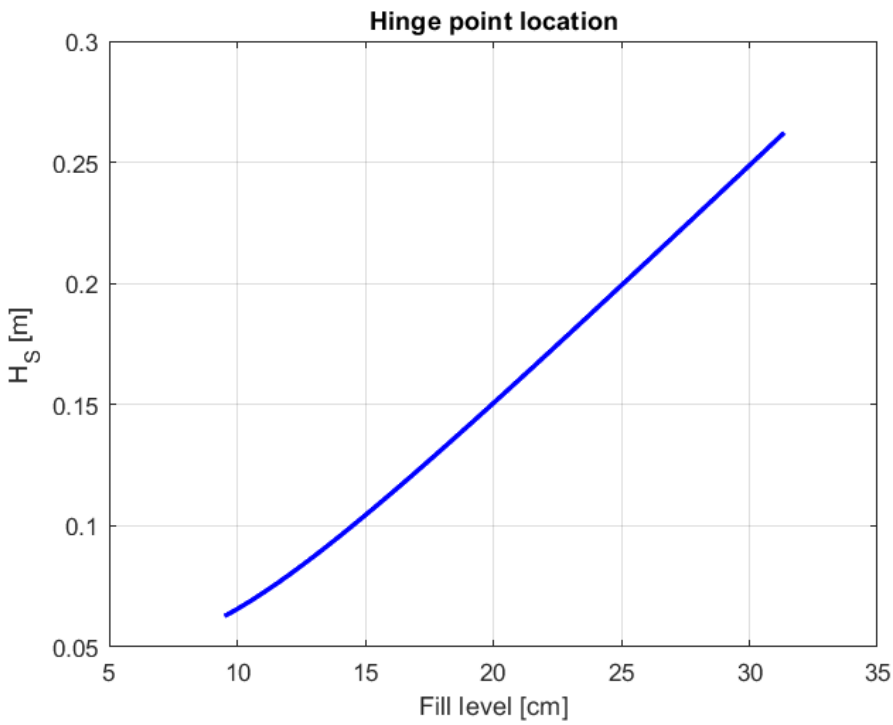


Figure 3.5: Efesto LOX tank - Hinge point location

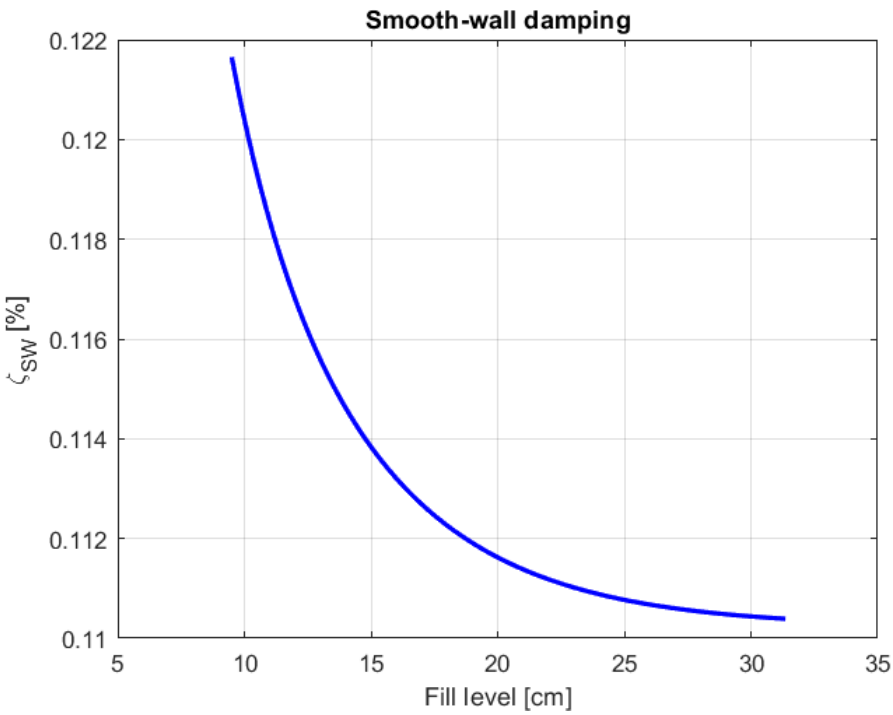


Figure 3.6: Efesto LOX tank - Smooth-wall damping

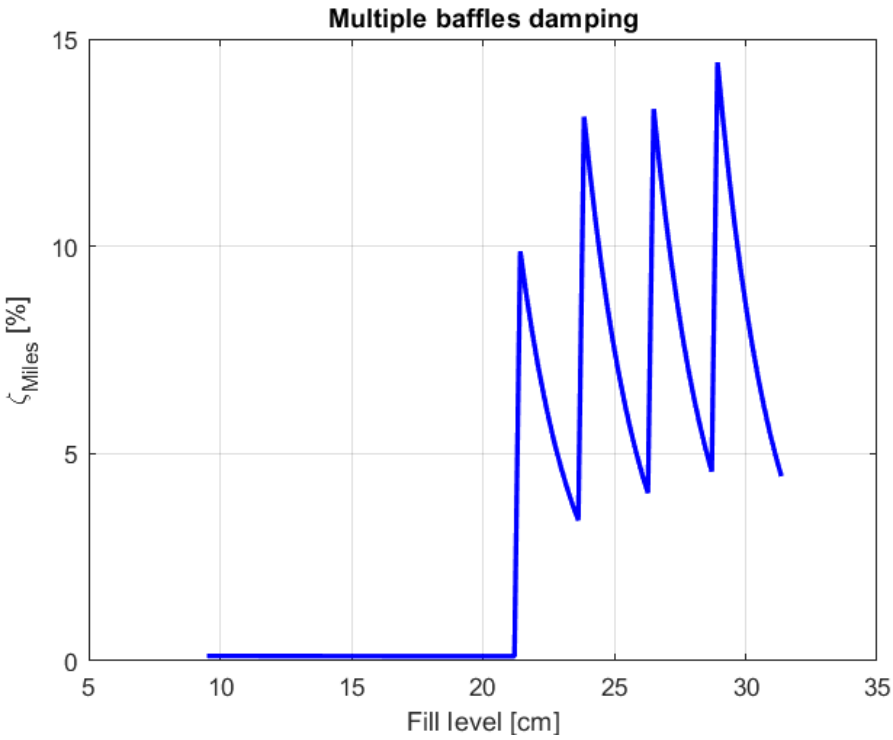


Figure 3.7: Efesto LOX tank - Multiple-baffle damping

All the curves show great accordance with the figures presented in subsection 2.2.2 as they are just scaled of some constant factor, except for the slosh mass.

Indeed, the dimensional form of slosh mass is obtained from the non-dimensional slosh mass equation (see table 2.1) by multiplying the total liquid mass, which is a function of fill level. This results in the graph of figure 3.4, where the slosh mass increases with fill level. What figure 2.4 describes, instead, is that the portion of the total mass which takes part in the sloshing is decreasing with fill level increasing.

As expected, multiple-baffle damping turns out to be at most two orders of magnitude bigger than the smooth-wall damping, again proving the efficacy of slosh-suppressing devices compared to the viscous damping of the fluid alone.

Free attenuation curve

The time evolution of the slosh mass displacement relative to its equilibrium point, $x = 0$, is computed for a given fill level of $h = 31.36\ cm$ from the bottom of the tank, which corresponds to the upper limit of the barrel section.

Using the equation of motion, 2.9, it is possible to plot the motion of the underdamped linear system as a function of time (figure 3.8). Given the maximum slosh wave amplitude for the Efesto tank, the initial displacement is set to be $x_0 = 9.1mm$, according to equation 2.14, whereas the initial velocity is zero.

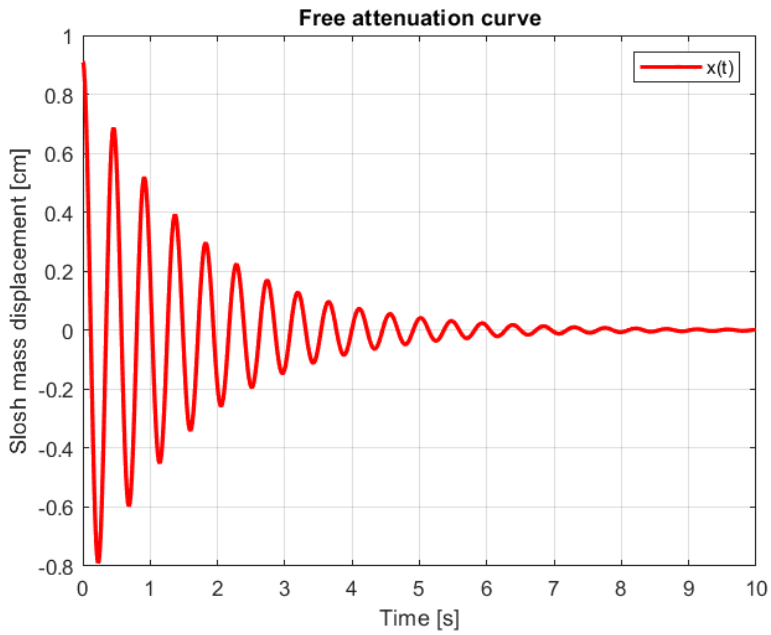


Figure 3.8: Free attenuation curve of Efesto LOX tank with four baffles

In order to provide a visual explanation of how the damping varies depending on the fill level, two other different cases are analyzed.
The first one features the highest damping value, which is obtained at the fill level relative to the #1 ring baffle, at $h = 28.88cm$ from the bottom of the tank. The other one simulates a bare-wall tank, namely the lowest damping value, and to avoid any interference with baffles the chosen fill level is the very beginning of the cylindrical section, at $h = 9.5\,cm$.

The free attenuation curves of said layouts are illustrated in figures 3.9 and 3.10, where the slosh mass displacement is evaluated over the same time span of 10 seconds. For coherent results, the initial conditions are the same as in the previous example.

	Fill level h [cm]	**Frequency** ω_0 [rad/s]	**Damping** ζ [%]
LOW damping	9.5	13.45	0.12
AVG damping	31.36	13.79	4.45
HIGH damping	28.88	13.79	14.80

Table 3.2: Comparison between different damping regimes at different fill levels

It is worth pointing out that these results are only approximations based on a linear mechanical model, which relies on a series of strong assumptions. Despite results may differ in real applications, the model succeeds in predicting the general trend pretty well. Indeed, as table 3.2 suggests, in the neighbourhood of a baffle the damping is quite intense, dropping significantly for high baffle depths or fill levels below the last baffle.

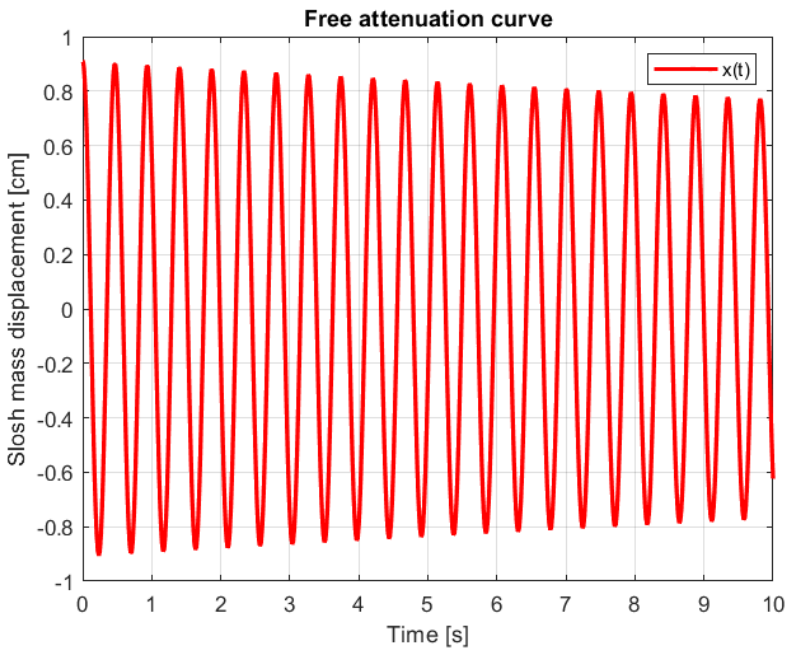


Figure 3.9: Free attenuation curve in the bare-wall configuration - lowest damping

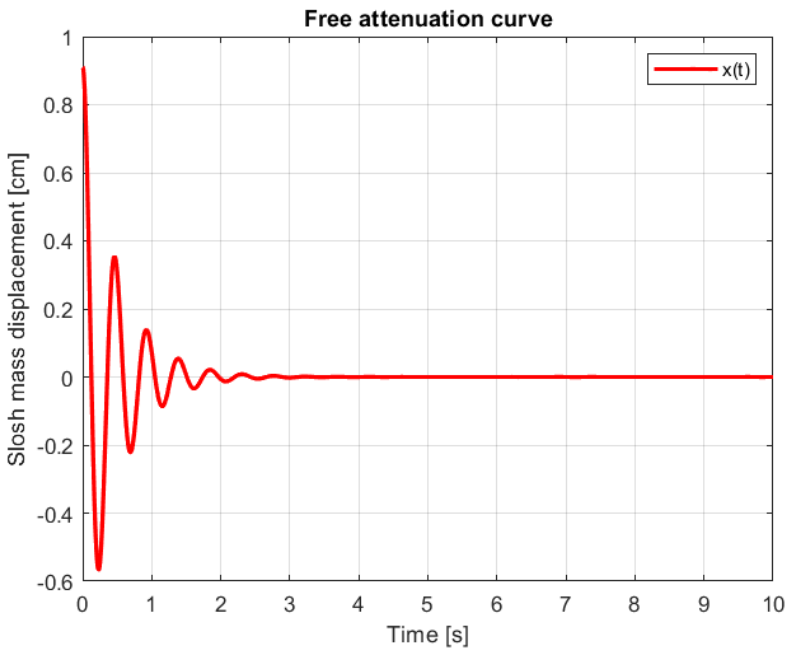


Figure 3.10: Free attenuation curve in the proximity of a baffle - highest damping

Chapter 4

Conclusions

This thesis mainly focused on evaluating how the sloshing in propellant tanks could be extinguished to avoid stability and controllability issues in space vehicles. Among several designs, the flat annular ring baffle was picked as the optimal device to suppress sloshing in liquid propellant tanks for rockets or satellites.

Such device has been scrupulously investigated through the use of a linear mechanical model, precisely a mass-pendulum-dashpot model, aiming at providing parameters which could properly describe the physical phenomenon. The mathematical expressions have been validated using experimental data collected by *Perez et al., 2012* [4] in the Ares I US LH2 tank model.

It is observed that baffle damping is independent of the working fluid at fill levels where the baffle contribution is dominant, unlike smooth-wall damping, where it varies upon the properties of the fluid. This explains the reason behind the need for a scaling factor when considering the damping of a bare tank.

Single-baffle and multiple-baffle configurations were studied and a discrete compliance with analytical results was found. The single-baffle test showed some discrepancies relative to the Miles equation, possibly due to inaccuracies when sample data were collected or to coefficients in the formula not being optimized for the tank geometry. Furthermore, Miles equation underestimates baffle damping at fill levels very close to the baffle location, as turbulent flow is there present, requiring more complicated non-linear analysis. In the multiple-baffle layout the linear superposition of Miles equation was proven accurate.

Since all the other mechanical model parameters demonstrated a substantial agreement with experimental results, the mechanical model has been implemented on a full-scale version of the *Efesto* liquid oxygen tank, by Polito Rocket Team. Here the damping provided by four baffles was calculated using the linear superposition of the Miles equation and utilized to compare the slosh mass displacement as a function of time for different damping regimes. Again, analysis pointed out the effectiveness of inner-mounted baffles to suppress liquid sloshing compared to a bare-wall tank relying purely on viscous damping.

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